Revision Notes

CBSE Class 12 Mathematics • NCERT Chapters 7–13

Integrals • Application of Integrals • Differential Equations • Vector Algebra
Three Dimensional Geometry • Linear Programming • Probability

How to use these notes. Keep the separate **formula sheet** next to you. Each chapter has four parts:

- **A JEE basics (extra)**: purple boxes; not needed for the board exam.
- **B Exam tips**: the "which method?" checklist, mistakes that lose marks, and question types.
- **C Exercise-wise solution patterns**: every NCERT exercise grouped into a few patterns, each with one solved question from the book. Try the solved question yourself first, then check. The small line under each exercise lists which questions belong to which pattern.
- **D Pattern spotter**: a one-look table for final revision. Cover the middle column and say the method aloud.

Question numbers match the NCERT textbook (Reprint 2026–27).

Contents

Chapter 7 Integrals **3**
7A JEE basics (extra) . . . 3
7B Exam tips . . . 4
7C Exercise-wise solution patterns . . . 5
7D Pattern spotter . . . 13

Chapter 8 Application of Integrals **16**
8A JEE basics (extra) . . . 16
8B Exam tips . . . 17
8C Exercise-wise solution patterns . . . 17
8D Pattern spotter . . . 18

Chapter 9 Differential Equations **19**
9A JEE basics (extra) . . . 19
9B Exam tips . . . 20
9C Exercise-wise solution patterns . . . 21
9D Pattern spotter . . . 24

Chapter 10 Vector Algebra **26**
10A JEE basics (extra) . . . 26
10B Exam tips . . . 27
10C Exercise-wise solution patterns . . . 28
10D Pattern spotter . . . 31

Chapter 11 Three Dimensional Geometry **32**

11A JEE basics (extra) . . . 32
11B Exam tips . . . 33
11C Exercise-wise solution patterns . . . 33
11D Pattern spotter . . . 35

Chapter 12 Linear Programming 36
12A JEE basics (extra) . . . 36
12B Exam tips . . . 36
12C Exercise-wise solution patterns . . . 37
12D Pattern spotter . . . 38

Chapter 13 Probability 39
13A JEE basics (extra) . . . 39
13B Exam tips . . . 39
13C Exercise-wise solution patterns . . . 40
13D Pattern spotter . . . 42

Chapter 7 Integrals

7A JEE basics (extra)

J1. Standard substitutions

$\sqrt{a^2 - x^2}$	$x = a\sin\theta$
$\sqrt{a^2 + x^2}$ or $a^2 + x^2$	$x = a\tan\theta$
$\sqrt{x^2 - a^2}$	$x = a\sec\theta$
$\sqrt{\dfrac{a-x}{a+x}}$	$x = a\cos 2\theta$
$\dfrac{1}{a+b\cos x}$, $\dfrac{1}{a+b\sin x}$	$t = \tan\dfrac{x}{2}$: $\sin x = \dfrac{2t}{1+t^2}$, $\cos x = \dfrac{1-t^2}{1+t^2}$, $dx = \dfrac{2\,dt}{1+t^2}$
$\dfrac{1}{a\sin^2 x + b\cos^2 x}$	divide by $\cos^2 x$, then $t = \tan x$
$\dfrac{a\sin x + b\cos x}{c\sin x + d\cos x}$	Numerator $= A\,(\text{Den}) + B\,(\text{Den})'$

J2. Useful results

$$\int e^{ax}\sin bx\,dx = \frac{e^{ax}}{a^2+b^2}(a\sin bx - b\cos bx) + C$$

$$\int e^{ax}\cos bx\,dx = \frac{e^{ax}}{a^2+b^2}(a\cos bx + b\sin bx) + C$$

Wallis: $\displaystyle\int_0^{\pi/2}\sin^n x\,dx = \int_0^{\pi/2}\cos^n x\,dx = \frac{n-1}{n}\cdot\frac{n-3}{n-2}\cdots\times\begin{cases}\frac{\pi}{2} & n \text{ even}\\ 1 & n \text{ odd}\end{cases}$

e.g. $\int_0^{\pi/2}\sin^4 x\,dx = \frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2} = \frac{3\pi}{16}$

Periodic: if f has period T, $\int_0^{nT} f(x)\,dx = n\int_0^T f(x)\,dx$.

Leibniz rule: $\dfrac{d}{dx}\displaystyle\int_{g(x)}^{h(x)} f(t)\,dt = f(h(x))\,h'(x) - f(g(x))\,g'(x)$

J3. Two classic tricks

(a) $\displaystyle\int\frac{x^2+1}{x^4+1}\,dx$: divide by x^2 to get $\displaystyle\int\frac{1+\frac{1}{x^2}}{\left(x-\frac{1}{x}\right)^2+2}\,dx$, put $t = x - \frac{1}{x}$:

$$= \frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{x^2-1}{\sqrt{2}\,x}\right) + C$$

(b) $\displaystyle\int(\sqrt{\tan x} + \sqrt{\cot x})\,dx = \int\frac{\sin x + \cos x}{\sqrt{\sin x\cos x}}\,dx$. Put $t = \sin x - \cos x$, so $\sin x\cos x = \frac{1-t^2}{2}$:

$$= \sqrt{2}\int\frac{dt}{\sqrt{1-t^2}} = \sqrt{2}\sin^{-1}(\sin x - \cos x) + C$$

7B Exam tips

C1. Which method should I use? (check in this order)

1. **Is it a standard form?** (see formula sheet) Write the answer directly.
2. **Can it be simplified?** Expand, split the fraction, or use a trig identity.
3. **Is a function and its derivative both present?** Use substitution.
4. **Is it a rational function (polynomial / polynomial)?**
   - Denominator factorises $\rightarrow$ partial fractions.
   - Denominator is a quadratic that does not factorise $\rightarrow$ complete the square (Type 1 / Type 2).
5. **Is it a product of two different kinds of function?** $\rightarrow$ by parts (ILATE). Look for the $e^x[f + f']$ pattern first.
6. **Definite integral with limits like 0 to a, or $-a$ to a?** Try properties P_4, P_6, P_7 *before* integrating.
7. **Modulus or greatest integer function?** Split the interval where the expression changes.

C2. Common mistakes that lose marks

- Forgetting $+C$ in indefinite integrals.
- Not **changing the limits** after substitution in a definite integral (or mixing old limits with the new variable).
- Writing $\log x$ instead of $\log|x|$.
- Using partial fractions on an improper fraction without dividing first.
- Sign errors: $\int \sin x\,dx = -\cos x$ and $\frac{d}{dx}\cos x = -\sin x$.
- Forgetting the factor when the inside is linear: $\int \cos 3x\,dx = \frac{\sin 3x}{3}$, not $\sin 3x$.
- In P_4 questions, forgetting to **add** the two forms of I to get $2I$.
- For $|f(x)|$, splitting at wrong points. Always find the roots of $f(x)$ first.
- Writing $\int \frac{1}{x^2+4}\,dx = \tan^{-1}\frac{x}{2}$ (missing the $\frac{1}{2}$).

C3. Smart exam habits

- **Check your answer by differentiating it.** You should get back the integrand.
- Write the substitution clearly: "Put $t = \ldots$, so $dt = \ldots$". Step marks are given for this.
- In partial fractions, show the values of A, B, C clearly.
- For property questions, state which property you used (e.g. "using $\int_0^a f(x)dx = \int_0^a f(a-x)dx$").
- Box the final answer.

C4. Question types to practise

- **Short / MCQ:** direct standard integrals, $\int e^x[f + f']dx$, odd/even function properties, simple definite integrals, $\frac{d}{dx}\int_a^x f(t)dt$.
- **Medium:** substitution, Type 1 / Type 2 special integrals, partial fractions, by parts.
- **Long:** definite integrals using P_4 (e.g. $\int_0^\pi \frac{x\sin x}{1+\cos^2 x}dx$), $\int_0^{\pi/2}\log\sin x\,dx$, modulus functions.
- **Must-do from NCERT:** all solved examples, the Miscellaneous Exercise, and the property-based examples in the last section.

7C Exercise-wise solution patterns

Exercise 7.1 (22 questions, 5 patterns)

Pattern A: Inspection (reverse differentiation)

Spot: "Find an anti derivative by inspection." **Method:** guess a function whose derivative gives the integrand, then fix the constant factor.

Solved (Q5): $\sin 2x - 4e^{3x}$. Since $\frac{d}{dx}(\cos 2x) = -2\sin 2x$ and $\frac{d}{dx}(e^{3x}) = 3e^{3x}$, anti derivative $= -\frac{1}{2}\cos 2x - \frac{4}{3}e^{3x}$.

Pattern B: Expand / split, then power rule term by term

Spot: a bracket squared, a product, or a sum of standard functions. **Method:** expand, write roots as powers ($\sqrt{x} = x^{1/2}$), integrate each term.

Solved (Q10): $$\int \left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)^2 dx = \int \left(x - 2 + \frac{1}{x}\right) dx = \frac{x^2}{2} - 2x + \log|x| + C$$

Pattern C: Divide by the denominator (or factorise and cancel)

Spot: polynomial ÷ monomial (x^2, $\sqrt{x}$), or a numerator that factorises.

Solved (Q13): $x^3 - x^2 + x - 1 = x^2(x-1) + (x-1) = (x-1)(x^2+1)$.
$$\int \frac{(x-1)(x^2+1)}{x-1} dx = \int (x^2+1) dx = \frac{x^3}{3} + x + C$$

Pattern D: Rewrite trig into standard derivatives

Spot: trig fractions like $\frac{\dots}{\cos^2 x}$. **Method:** split and convert to $\sec^2 x$, $\sec x \tan x$, etc.

Solved (Q20): $$\int \frac{2 - 3\sin x}{\cos^2 x} dx = \int (2\sec^2 x - 3\sec x \tan x)\, dx = 2\tan x - 3\sec x + C$$

Pattern E: Use a given condition to find C

Solved (Q22): $f'(x) = 4x^3 - 3x^{-4} \Rightarrow f(x) = x^4 + x^{-3} + C$.
$f(2) = 0 \Rightarrow 16 + \frac{1}{8} + C = 0 \Rightarrow C = -\frac{129}{8}$. So $f(x) = x^4 + \frac{1}{x^3} - \frac{129}{8}$ **(A)**.

Pattern A: Q1–5. Pattern B: Q6–10, 14–17, 21. Pattern C: Q11–13. Pattern D: Q18–20. Pattern E: Q22.

Exercise 7.2 (39 questions, 5 patterns)

Pattern A: Numerator is the derivative of the denominator $\Rightarrow \log|f(x)|$

Spot: differentiate the denominator; do you get the numerator (up to a constant)?

Solved (Q24): $\int \frac{2\cos x - 3\sin x}{6\cos x + 4\sin x} dx$. $\frac{d}{dx}(6\cos x + 4\sin x) = 4\cos x - 6\sin x = 2(2\cos x - 3\sin x)$.
Put $t = 6\cos x + 4\sin x$, $dt = 2(2\cos x - 3\sin x)dx$: $\frac{1}{2}\int \frac{dt}{t} = \frac{1}{2}\log|6\cos x + 4\sin x| + C$

Pattern B: Power of $f(x)$ times $f'(x)$ $\Rightarrow$ put $t = f(x)$, then power rule

Solved (Q9): $\int (4x+2)\sqrt{x^2+x+1}\, dx$. Put $t = x^2+x+1$, $dt = (2x+1)dx$, and $4x+2 = 2(2x+1)$.

$= 2\int t^{1/2}dt = \frac{4}{3}(x^2+x+1)^{3/2} + C$

Pattern C: Function of a function, $f(g(x))\cdot g'(x)$ $\Rightarrow$ put $t = g(x)$

Solved (Q37): $\int \frac{x^3\sin(\tan^{-1}x^4)}{1+x^8}dx$. Put $t = x^4$, $dt = 4x^3dx$: $\frac{1}{4}\int \frac{\sin(\tan^{-1}t)}{1+t^2}dt$.
Now put $u = \tan^{-1}t$, $du = \frac{dt}{1+t^2}$: $\frac{1}{4}\int \sin u\,du = -\frac{1}{4}\cos(\tan^{-1}x^4) + C$

Pattern D: Linear expression inside + extra x outside $\Rightarrow$ put $t =$ inside, write x in terms of t

Solved (Q7): $\int x\sqrt{x+2}\,dx$. Put $t = x+2$, so $x = t-2$, $dx = dt$.
$= \int (t-2)t^{1/2}dt = \frac{2}{5}t^{5/2} - \frac{4}{3}t^{3/2} = \frac{2}{5}(x+2)^{5/2} - \frac{4}{3}(x+2)^{3/2} + C$

Pattern E: Trig rewrite first (identity or "split the numerator" trick)

Solved (Q33): $\int \frac{dx}{1-\tan x} = \int \frac{\cos x}{\cos x - \sin x}dx$. Write $\cos x = \frac{1}{2}\big[(\cos x - \sin x) + (\cos x + \sin x)\big]$:
$= \frac{1}{2}\int dx + \frac{1}{2}\int \frac{\cos x + \sin x}{\cos x - \sin x}dx = \frac{x}{2} - \frac{1}{2}\log|\cos x - \sin x| + C$
(The second integral is Pattern A: derivative of $\cos x - \sin x$ is $-(\sin x + \cos x)$.)

Pattern A: Q1, 3, 10, 15, 19, 20, 24, 30, 38. Pattern B: Q2, 5, 8, 9, 13, 14, 23, 25, 27–29, 31, 35, 36. Pattern C: Q4, 16–18, 22, 26, 37. Pattern D: Q6, 7, 11, 12. Pattern E: Q21, 32–34, 39.

Exercise 7.3 (24 questions, 5 patterns)

Pattern A: Even powers $\sin^2, \cos^2, \sin^4, \cos^4$ $\Rightarrow$ power-reduction formulas

Solved (Q10): $\sin^4 x = \left(\frac{1-\cos 2x}{2}\right)^2 = \frac{1}{4}\left(1 - 2\cos 2x + \frac{1+\cos 4x}{2}\right) = \frac{3}{8} - \frac{1}{2}\cos 2x + \frac{1}{8}\cos 4x$.
$\int \sin^4 x\,dx = \frac{3x}{8} - \frac{\sin 2x}{4} + \frac{\sin 4x}{32} + C$

Pattern B: Products like $\sin A\cos B$ $\Rightarrow$ product-to-sum formulas

Solved (Q3): $\cos 2x\cos 4x = \frac{1}{2}(\cos 6x + \cos 2x)$, so
$\cos 2x\cos 4x\cos 6x = \frac{1}{2}(\cos^2 6x + \cos 2x\cos 6x) = \frac{1}{4}\big[1 + \cos 12x + \cos 8x + \cos 4x\big]$.
$\int = \frac{1}{4}\left[x + \frac{\sin 12x}{12} + \frac{\sin 8x}{8} + \frac{\sin 4x}{4}\right] + C$

Pattern C: Odd powers $\Rightarrow$ keep one factor aside, convert the rest, substitute

Solved (Q5): $\sin^3 x\cos^3 x = \sin^3 x(1-\sin^2 x)\cos x$. Put $t = \sin x$, $dt = \cos x\,dx$:
$\int (t^3 - t^5)dt = \frac{\sin^4 x}{4} - \frac{\sin^6 x}{6} + C$

Pattern D: Simplify the fraction using an identity

Tools: $1 \pm \cos x = 2\cos^2\frac{x}{2}$ or $2\sin^2\frac{x}{2}$, $1 + \sin 2x = (\sin x + \cos x)^2$, $\cos 2x = \cos^2 x - \sin^2 x$.

Solved (Q8): $\dfrac{1-\cos x}{1+\cos x} = \dfrac{2\sin^2\frac{x}{2}}{2\cos^2\frac{x}{2}} = \tan^2\frac{x}{2} = \sec^2\frac{x}{2} - 1.$ $\displaystyle\int = 2\tan\frac{x}{2} - x + C$

Pattern E: Special tricks

Solved (Q22): $\displaystyle\int \frac{dx}{\cos(x-a)\cos(x-b)}$. Multiply and divide by $\sin(a-b) = \sin\big[(x-b)-(x-a)\big]$:

$$\frac{\sin[(x-b)-(x-a)]}{\cos(x-a)\cos(x-b)} = \tan(x-b) - \tan(x-a).$$

$$\int = \frac{1}{\sin(a-b)}\Big[-\log|\cos(x-b)| + \log|\cos(x-a)|\Big] = \frac{1}{\sin(a-b)}\log\left|\frac{\cos(x-a)}{\cos(x-b)}\right| + C$$

Pattern A: Q1, 10, 11. Pattern B: Q2, 3, 6, 7. Pattern C: Q4, 5, 15, 16. Pattern D: Q8, 9, 12–14, 17, 18, 20, 23. Pattern E: Q19, 21, 22, 24.

Exercise 7.4 (25 questions, 4 patterns)

Pattern A: Already a standard form (just adjust the coefficient of x^2)

Solved (Q4): $\displaystyle\int \frac{dx}{\sqrt{9-25x^2}} = \frac{1}{5}\int \frac{dx}{\sqrt{(3/5)^2 - x^2}} = \frac{1}{5}\sin^{-1}\frac{5x}{3} + C$

Pattern B: Substitute to reach a standard form ($t = x^2$, x^3, $\tan x, \ldots$)

Solved (Q6): $\displaystyle\int \frac{x^2}{1-x^6}dx$. Put $t = x^3$, $dt = 3x^2dx$: $\displaystyle\frac{1}{3}\int \frac{dt}{1-t^2} = \frac{1}{3}\cdot\frac{1}{2}\log\left|\frac{1+t}{1-t}\right| = \frac{1}{6}\log\left|\frac{1+x^3}{1-x^3}\right| + C$

Pattern C: Quadratic in the denominator, constant numerator $\Rightarrow$ complete the square

Solved (Q12): $7 - 6x - x^2 = -(x^2+6x-7) = 16 - (x+3)^2$. $\displaystyle\int \frac{dx}{\sqrt{16-(x+3)^2}} = \sin^{-1}\frac{x+3}{4} + C$

Pattern D: Linear numerator $\Rightarrow px + q = A\cdot\frac{d}{dx}(\text{quadratic}) + B$

Solved (Q18): $\displaystyle\int \frac{5x-2}{3x^2+2x+1}dx$. $\frac{d}{dx}(3x^2+2x+1) = 6x+2$, and $5x - 2 = \frac{5}{6}(6x+2) - \frac{11}{3}$.

$$= \frac{5}{6}\log|3x^2+2x+1| - \frac{11}{3}\int \frac{dx}{3\left[(x+\frac{1}{3})^2 + (\frac{\sqrt{2}}{3})^2\right]} = \frac{5}{6}\log|3x^2+2x+1| - \frac{11}{3\sqrt{2}}\tan^{-1}\frac{3x+1}{\sqrt{2}} + C$$

Pattern A: Q2, 4. Pattern B: Q1, 3, 5, 6, 8, 9. Pattern C: Q10–15, 24, 25. Pattern D: Q7, 16–23.

Exercise 7.5 (23 questions, 5 patterns)

Pattern A: Distinct linear factors

Solved (Q3): $\dfrac{3x-1}{(x-1)(x-2)(x-3)} = \dfrac{A}{x-1}+\dfrac{B}{x-2}+\dfrac{C}{x-3}$, so $3x-1 = A(x-2)(x-3)+B(x-1)(x-3)+C(x-1)(x-2)$.

$x=1$: $2=2A \Rightarrow A=1$; $x=2$: $5=-B \Rightarrow B=-5$; $x=3$: $8=2C \Rightarrow C=4$.

$$\int = \log|x-1| - 5\log|x-2| + 4\log|x-3| + C$$

Pattern B: Repeated linear factor $(x-a)^2$

Solved (Q14): $\dfrac{3x-1}{(x+2)^2} = \dfrac{A}{x+2}+\dfrac{B}{(x+2)^2}$, so $3x-1 = A(x+2)+B \Rightarrow A=3,\ B=-7$.

$$\int = 3\log|x+2| + \frac{7}{x+2} + C$$

Pattern C: A quadratic factor that cannot be factorised $\Rightarrow \dfrac{Bx+C}{x^2+\ldots}$

Solved (Q13): $\dfrac{2}{(1-x)(1+x^2)} = \dfrac{A}{1-x}+\dfrac{Bx+C}{1+x^2}$, so $2 = A(1+x^2)+(Bx+C)(1-x)$.

$x=1$: $A=1$; coefficient of x^2: $A-B=0 \Rightarrow B=1$; constant: $A+C=2 \Rightarrow C=1$.

$$\int = -\log|1-x| + \frac{1}{2}\log(1+x^2) + \tan^{-1}x + C$$

Pattern D: Improper fraction $\Rightarrow$ divide first

Solved (Q12): $\dfrac{x^3+x+1}{x^2-1} = x + \dfrac{2x+1}{(x-1)(x+1)} = x + \dfrac{3/2}{x-1} + \dfrac{1/2}{x+1}$.

$$\int = \frac{x^2}{2} + \frac{3}{2}\log|x-1| + \frac{1}{2}\log|x+1| + C$$

Pattern E: Substitute first, then partial fractions ($t=\sin x,\ e^x,\ x^2,\ x^n$)

Solved (Q17): $\displaystyle\int \frac{\cos x}{(1-\sin x)(2-\sin x)}dx$. Put $t=\sin x$: $\displaystyle\int \frac{dt}{(1-t)(2-t)} = \int\left(\frac{1}{1-t}-\frac{1}{2-t}\right)dt$

$$= -\log|1-t| + \log|2-t| = \log\left|\frac{2-\sin x}{1-\sin x}\right| + C$$

Pattern A: Q1–5, 10, 11, 22. Pattern B: Q8, 9, 14. Pattern C: Q7, 13, 15, 23. Pattern D: Q6, 12. Pattern E: Q16–21.

Exercise 7.6 (24 questions, 5 patterns)

Pattern A: Algebraic × trig / exponential ⇒ u = algebraic (repeat if needed)

Solved (Q3): $\int x^2 e^x dx = x^2 e^x - \int 2xe^x dx = x^2 e^x - 2(xe^x - e^x) = e^x(x^2 - 2x + 2) + C$

Pattern B: Log × algebraic ⇒ $u = \log x$

Solved (Q6): $\int x^2 \log x \, dx = \log x \cdot \frac{x^3}{3} - \int \frac{1}{x} \cdot \frac{x^3}{3} dx = \frac{x^3}{3} \log x - \frac{x^3}{9} + C$

Pattern C: Inverse trig ⇒ u = inverse trig (use 1 as the second function if alone)

Solved (Q8): $\int x \tan^{-1} x \, dx = \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int \frac{x^2}{1+x^2} dx = \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int \left(1 - \frac{1}{1+x^2}\right) dx$

$= \frac{x^2+1}{2} \tan^{-1} x - \frac{x}{2} + C$

Pattern D: $e^x[f(x) + f'(x)]$ ⇒ $e^x f(x)$ (rearrange the bracket to see f and f')

Solved (Q17): $\frac{x}{(1+x)^2} = \frac{(1+x)-1}{(1+x)^2} = \frac{1}{1+x} - \frac{1}{(1+x)^2}$. Here $f = \frac{1}{1+x}$, $f' = -\frac{1}{(1+x)^2}$.

$\int \frac{xe^x}{(1+x)^2} dx = \frac{e^x}{1+x} + C$

Pattern E: The integral comes back (cyclic) ⇒ by parts twice, solve for I

Solved (Q21): $I = \int e^{2x} \sin x \, dx = \frac{e^{2x}}{2} \sin x - \frac{1}{2} \int e^{2x} \cos x \, dx = \frac{e^{2x}}{2} \sin x - \frac{1}{2} \left[\frac{e^{2x}}{2} \cos x + \frac{1}{2} I\right]$

$\Rightarrow \frac{5}{4} I = \frac{e^{2x}}{4}(2 \sin x - \cos x) \Rightarrow I = \frac{e^{2x}}{5}(2 \sin x - \cos x) + C$

Pattern A: Q1, 2, 3, 12. Pattern B: Q4, 5, 6, 14, 15. Pattern C: Q7–11, 13, 22. Pattern D: Q16–20, 24. Pattern E: Q21. (Q23 is substitution, not by parts.)

Exercise 7.7 (11 questions, 2 patterns)

Pattern A: Direct $\sqrt{a^2 \pm x^2}$ or $\sqrt{x^2 - a^2}$ (take out the coefficient of x^2 first)

Solved (Q2): $\sqrt{1 - 4x^2} = 2\sqrt{(\frac{1}{2})^2 - x^2}$.

$\int = 2\left[\frac{x}{2}\sqrt{\frac{1}{4} - x^2} + \frac{1}{8} \sin^{-1} 2x\right] = \frac{x}{2}\sqrt{1 - 4x^2} + \frac{1}{4} \sin^{-1} 2x + C$

Pattern B: $\sqrt{\text{quadratic}}$ ⇒ complete the square, then the $\sqrt{}$ formula

Solved (Q6): $x^2 + 4x - 5 = (x+2)^2 - 3^2$.

$\int = \frac{x+2}{2}\sqrt{x^2 + 4x - 5} - \frac{9}{2} \log \left|x + 2 + \sqrt{x^2 + 4x - 5}\right| + C$

Pattern A: Q1, 2, 9, 10. Pattern B: Q3–8, 11.

Exercise 7.8 (22 questions, 2 patterns)

Pattern A: Direct antiderivative, then $F(b) - F(a)$

Solved (Q8): $\displaystyle\int_{\pi/6}^{\pi/4} \operatorname{cosec} x\, dx = \Big[\log|\operatorname{cosec} x - \cot x|\Big]_{\pi/6}^{\pi/4} = \log(\sqrt{2}-1) - \log(2-\sqrt{3}) = \log\frac{\sqrt{2}-1}{2-\sqrt{3}}$

Pattern B: Use a method first (identity / substitution / special form / partial fractions / parts)

Solved (Q16): $\displaystyle\frac{5x^2}{x^2+4x+3} = 5 - \frac{20x+15}{(x+1)(x+3)} = 5 + \frac{5/2}{x+1} - \frac{45/2}{x+3}.$

$\displaystyle\int_1^2 = \Big[5x + \tfrac{5}{2}\log|x+1| - \tfrac{45}{2}\log|x+3|\Big]_1^2 = 5 + \frac{5}{2}\log\frac{3}{2} - \frac{45}{2}\log\frac{5}{4}$

Pattern A: Q1–10, 17, 21, 22. Pattern B: Q11–16, 18–20.

Exercise 7.9 (10 questions, 4 patterns)

Pattern A: Substitute and change the limits

Solved (Q5): $\displaystyle\int_0^{\pi/2} \frac{\sin x}{1+\cos^2 x}dx$. Put $t = \cos x$, $dt = -\sin x\, dx$; $x = 0 \to t = 1$, $x = \frac{\pi}{2} \to t = 0$.

$\displaystyle= \int_1^0 \frac{-dt}{1+t^2} = \int_0^1 \frac{dt}{1+t^2} = \frac{\pi}{4}$

Pattern B: Trig substitution to simplify an inverse-trig expression

Solved (Q3): Put $x = \tan\theta$: $\sin^{-1}\frac{2\tan\theta}{1+\tan^2\theta} = \sin^{-1}(\sin 2\theta) = 2\theta$, $dx = \sec^2\theta\, d\theta$, limits $0 \to \frac{\pi}{4}$.

$\displaystyle\int_0^{\pi/4} 2\theta\sec^2\theta\, d\theta = 2\Big[\theta\tan\theta - \log|\sec\theta|\Big]_0^{\pi/4} = 2\left[\frac{\pi}{4} - \log\sqrt{2}\right] = \frac{\pi}{2} - \log 2$

Pattern C: Complete the square, then standard form with limits

Solved (Q7): $\displaystyle\int_{-1}^{1} \frac{dx}{(x+1)^2+2^2} = \left[\frac{1}{2}\tan^{-1}\frac{x+1}{2}\right]_{-1}^{1} = \frac{1}{2}\cdot\frac{\pi}{4} - 0 = \frac{\pi}{8}$

Pattern D: Substitute, then $e^t[f + f']$

Solved (Q8): Put $t = 2x$: $\displaystyle\int_1^2 \left(\frac{1}{x} - \frac{1}{2x^2}\right)e^{2x}dx = \int_2^4 e^t\left(\frac{1}{t} - \frac{1}{t^2}\right)dt = \left[\frac{e^t}{t}\right]_2^4 = \frac{e^4}{4} - \frac{e^2}{2}$

Pattern A: Q1, 2, 4, 5, 9. Pattern B: Q3. Pattern C: Q6, 7. Pattern D: Q8. Q10: first fundamental theorem.

Exercise 7.10 (21 questions, 5 patterns)

Pattern A: P_4 and add $\Rightarrow 2I = \int 1\,dx$

Spot: limits 0 to a; replacing x by $a-x$ swaps sin $\leftrightarrow$ cos (or $x \leftrightarrow a-x$).

Solved (Q4): $I = \displaystyle\int_0^{\pi/2} \frac{\cos^5 x}{\sin^5 x + \cos^5 x}dx$. By P_4: $I = \displaystyle\int_0^{\pi/2} \frac{\sin^5 x}{\cos^5 x + \sin^5 x}dx$. Add: $2I = \displaystyle\int_0^{\pi/2} 1\,dx = \frac{\pi}{2} \Rightarrow I = \frac{\pi}{4}$

Pattern B: P_4 to simplify or remove the x

Solved (Q8): $I = \displaystyle\int_0^{\pi/4} \log(1+\tan x)\,dx$. By P_4: $1+\tan(\tfrac{\pi}{4}-x) = 1 + \frac{1-\tan x}{1+\tan x} = \frac{2}{1+\tan x}$.

So $I = \displaystyle\int_0^{\pi/4} \big[\log 2 - \log(1+\tan x)\big]dx = \frac{\pi}{4}\log 2 - I \;\Rightarrow\; I = \frac{\pi}{8}\log 2$

Pattern C: P_4 turns the integrand into its negative $\Rightarrow I = -I \Rightarrow I = 0$

Solved (Q15): $I = \displaystyle\int_0^{\pi/2} \frac{\sin x - \cos x}{1+\sin x\cos x}dx$. By P_4: $I = \displaystyle\int_0^{\pi/2} \frac{\cos x - \sin x}{1+\cos x\sin x}dx = -I \Rightarrow I = 0$

Pattern D: Odd / even (P_7) and P_6

Solved (Q20): x^3, $x\cos x$, $\tan^5 x$ are odd, so their integrals over $[-\frac{\pi}{2}, \frac{\pi}{2}]$ are 0. Remaining $\displaystyle\int_{-\pi/2}^{\pi/2} 1\,dx = \pi$ **(C)**.

Pattern E: Modulus $\Rightarrow$ split where the inside is zero

Solved (Q6): $|x-5|$ changes at $x=5$: $\displaystyle\int_2^5 (5-x)\,dx + \int_5^8 (x-5)\,dx = \frac{9}{2} + \frac{9}{2} = 9$

Pattern A: Q1–4, 17. Pattern B: Q7–10, 12, 16, 19. Pattern C: Q15, 21. Pattern D: Q11, 13, 14, 20. Pattern E: Q5, 6, 18.

Miscellaneous Exercise (40 questions, 7 patterns)

These mix all earlier patterns. Spotting the pattern is the main skill.

Pattern A: Partial fractions (see Ex 7.5)

Solved (Q14): Put $x^2 = y$ to find constants: $\dfrac{1}{(y+1)(y+4)} = \frac{1}{3}\left[\frac{1}{y+1} - \frac{1}{y+4}\right]$.

$\displaystyle\int \frac{dx}{(x^2+1)(x^2+4)} = \frac{1}{3}\tan^{-1}x - \frac{1}{6}\tan^{-1}\frac{x}{2} + C$

Pattern B: Rationalise (multiply by the conjugate)

Solved (Q28): $\dfrac{1}{\sqrt{1+x}-\sqrt{x}} = \sqrt{1+x}+\sqrt{x}$ (denominator becomes 1).

$\displaystyle\int_0^1 (\sqrt{1+x}+\sqrt{x})\,dx = \frac{2}{3}\Big[(1+x)^{3/2} + x^{3/2}\Big]_0^1 = \frac{2}{3}(2\sqrt{2}+1-1) = \frac{4\sqrt{2}}{3}$

Pattern C: Simplify first (exp–log rules, trig identities)

Solved (Q10): $\sin^8 x - \cos^8 x = (\sin^4 x - \cos^4 x)(\sin^4 x + \cos^4 x) = (\sin^2 x - \cos^2 x)(\sin^4 x + \cos^4 x)$, and $1 - 2\sin^2 x \cos^2 x = \sin^4 x + \cos^4 x$. So integrand $= \sin^2 x - \cos^2 x = -\cos 2x$. $\displaystyle\int = -\frac{1}{2}\sin 2x + C$

Pattern D: Smart substitution (take out the highest power; $t = \sin x - \cos x$; hint substitutions)

Solved (Q4): $x^2(x^4+1)^{3/4} = x^2 \cdot x^3 \left(1 + \frac{1}{x^4}\right)^{3/4} = x^5 \left(1 + \frac{1}{x^4}\right)^{3/4}$.

Put $t = 1 + \frac{1}{x^4}$, $dt = -\frac{4}{x^5}dx$: $\displaystyle -\frac{1}{4}\int t^{-3/4} dt = -t^{1/4} = -\left(1 + \frac{1}{x^4}\right)^{1/4} + C$

Solved (Q29): Put $t = \sin x - \cos x$, $dt = (\cos x + \sin x)dx$, and $\sin 2x = 1 - t^2$; limits $-1 \to 0$.

$$\int_{-1}^{0} \frac{dt}{25 - 16t^2} = \frac{1}{40}\left[\log\left|\frac{5+4t}{5-4t}\right|\right]_{-1}^{0} = 0 - \frac{1}{40}\log\frac{1}{9} = \frac{1}{20}\log 3$$

Pattern E: Trig substitution for roots / inverse trig

Solved (Q22): Put $x = \cos 2\theta$: $\sqrt{\frac{1-\cos 2\theta}{1+\cos 2\theta}} = \tan\theta$, so $\tan^{-1}(\tan\theta) = \theta = \frac{1}{2}\cos^{-1} x$.

$$\int \frac{1}{2}\cos^{-1} x\, dx = \frac{1}{2}\left[x\cos^{-1} x - \sqrt{1-x^2}\right] + C \quad \text{(by parts with 1 as the second function)}$$

Pattern F: By parts / $e^x[f + f']$

Solved (Q24): $\displaystyle\frac{1-\sin x}{1-\cos x} = \frac{1 - 2\sin\frac{x}{2}\cos\frac{x}{2}}{2\sin^2\frac{x}{2}} = \frac{1}{2}\text{cosec}^2\frac{x}{2} - \cot\frac{x}{2}$. So $f = -\cot\frac{x}{2}$, $f' = \frac{1}{2}\text{cosec}^2\frac{x}{2}$.

$$\int_{\pi/2}^{\pi} e^x(f + f')\, dx = \left[-e^x \cot\tfrac{x}{2}\right]_{\pi/2}^{\pi} = 0 - (-e^{\pi/2} \cdot 1) = e^{\pi/2}$$

Pattern G: Properties of definite integrals / modulus

Solved (Q31): Find each part separately on $[1, 4]$:

$\int_1^4 |x-1| = \frac{9}{2}$, $\int_1^4 |x-2| = \frac{1}{2} + 2 = \frac{5}{2}$, $\int_1^4 |x-3| = 2 + \frac{1}{2} = \frac{5}{2}$. Total $= \dfrac{19}{2}$

Pattern A: Q1, 6, 13, 14, 21, 32. Pattern B: Q2, 28. Pattern C: Q8, 10, 11, 15, 16, 35, 36, 39. Pattern D: Q3–5, 7, 9, 12, 17, 18, 23, 25–27, 29, 38. Pattern E: Q19, 22. Pattern F: Q20, 24, 30, 33, 37. Pattern G: Q31, 34, 40.

7D Pattern spotter

If you see...	Do this	See
1. Before any method: simplify		
A bracket squared, or a product of simple terms	Expand, write roots as powers, integrate term by term	7.1 B
Polynomial ÷ single term (x^2, $\sqrt{x}$)	Divide each term separately	7.1 C
Numerator factorises with the denominator	Factorise and cancel	7.1 C
$e^{k\log x}$, $e^{\log f(x)}$	Replace by x^k, $f(x)$	Misc C
$\dfrac{1}{\sqrt{A}\pm\sqrt{B}}$	Multiply by the conjugate (rationalise)	Misc B
2. Substitution		
Numerator = derivative of denominator (up to a constant)	$\log\lvert\text{denominator}\rvert$	7.2 A
$[f(x)]^n\cdot f'(x)$	$t=f(x)$, power rule	7.2 B
$f\big(g(x)\big)\cdot g'(x)$, e.g. $\frac{e^{\tan^{-1}x}}{1+x^2}$, $\frac{\cos\sqrt{x}}{\sqrt{x}}$	$t=g(x)$	7.2 C
$x\sqrt{x+a}$, $\dfrac{x}{\sqrt{x+a}}$ (linear inside, extra x)	$t=x+a$, write $x=t-a$	7.2 D
$\dfrac{e^x-e^{-x}}{e^x+e^{-x}}$ or $\dfrac{e^{2x}-1}{e^{2x}+1}$	Make numerator the derivative of the denominator	7.2 A
$x^{\text{small}}(x^{\text{big}}\pm1)^p$ with fractional p	Take the highest power of x out, then $t=1\pm x^{-n}$	Misc D
$\sin x\pm\cos x$ in the numerator, $\sin 2x$ in the denominator	$t=\sin x\mp\cos x$, then $\sin 2x=1-t^2$ or t^2-1	Misc D
3. Trigonometric integrals		
$\sin^2$, $\cos^2$, $\sin^4$, $\cos^4$	Power-reduction: $\sin^2\theta=\frac{1-\cos 2\theta}{2}$	7.3 A
$\sin A\cos B$, $\cos A\cos B$, $\sin A\sin B$	Product-to-sum formulas	7.3 B
Odd power ($\sin^3$, $\cos^5$, $\tan^3\sec$)	Keep one factor, convert the rest, substitute	7.3 C
$\tan^2$, $\tan^4$	$\tan^2=\sec^2-1$ (repeat)	7.3 C
$1\pm\cos x$, $1\pm\sin 2x$, $\cos 2x$ in a fraction	Half-angle / $(\sin x\pm\cos x)^2$ / factorise $\cos 2x$	7.3 D
$\dfrac{1}{\sin^2x\cos^2x}$, $\dfrac{\sin^2x-\cos^2x}{\sin^2x\cos^2x}$	Split using $1=\sin^2+\cos^2$	7.2 E
$\dfrac{1}{1\pm\tan x}$, $\dfrac{1}{1\pm\cot x}$	Convert to sin / cos; split numerator as $\frac{1}{2}[\text{Den}\pm\text{Den}']$	7.2 E

If you see...	Do this	See
$\frac{1}{\cos(x-a)\cos(x-b)}$ (or $\sin\cdot\sin$, $\sin\cdot\cos$)	Multiply and divide by $\sin(a-b)$ (or $\cos(a-b)$)	7.3 E
$\frac{1}{\sin x\cos^3 x}$, $\frac{\sqrt{\tan x}}{\sin x\cos x}$	Divide by a power of $\cos x$ to get $\sec^2 x$; $t=\tan x$	7.3 E
$\sin^{-1}(\cos x)$, $\cos^{-1}(\sin x)$	Convert: $\sin^{-1}(\cos x)=\frac{\pi}{2}-x$	7.3 E
4. Special standard forms		
$\frac{1}{x^2\pm a^2}$, $\frac{1}{a^2-x^2}$, $\frac{1}{\sqrt{\cdots}}$	Direct special-integral formula; first take out the coefficient of x^2	7.4 A
$\frac{x^2}{1\pm x^6}$, $\frac{\sec^2 x}{\tan^2 x+4}$, $\frac{x}{1+x^4}$	Substitute ($t=x^3,\ \tan x,\ x^2$) to reach a standard form	7.4 B
$\frac{1}{ax^2+bx+c}$ or $\frac{1}{\sqrt{ax^2+bx+c}}$	Complete the square	7.4 C
$\frac{px+q}{ax^2+bx+c}$ or $\frac{px+q}{\sqrt{ax^2+bx+c}}$	$px+q=A(2ax+b)+B$; split into log/$\sqrt{}$ part + standard part	7.4 D
$\sqrt{a^2-x^2}$, $\sqrt{x^2\pm a^2}$	$\sqrt{}$ formulas (formula sheet)	7.7 A
$\sqrt{ax^2+bx+c}$	Complete the square, then the $\sqrt{}$ formula	7.7 B
5. Partial fractions		
Rational function, degree of top $\geq$ degree of bottom	Divide first	7.5 D
Distinct linear factors	$\frac{A}{x-a}+\frac{B}{x-b}+\cdots$; put $x=a,\ b$	7.5 A
Repeated factor $(x-a)^2$	$\frac{A}{x-a}+\frac{B}{(x-a)^2}$	7.5 B
Quadratic factor that won't factorise	$\frac{Bx+C}{x^2+\cdots}$; compare coefficients	7.5 C
Only x^2 appears, e.g. $\frac{1}{(x^2+1)(x^2+4)}$	Put $x^2=y$ *just to find constants*	Misc A
Function of $\sin x$, e^x, x^n in factored denominator	Substitute first ($t=\sin x,\ e^x,\ x^n$), then partial fractions	7.5 E
6. Integration by parts		
Algebraic $\times$ trig or exponential	$u=$ algebraic (repeat for x^2)	7.6 A
$\log x$ times algebraic, or $\log x$ alone	$u=\log x$ (use 1 as v if alone)	7.6 B
Inverse trig times algebraic, or alone	$u=$ inverse trig (use 1 as v if alone)	7.6 C
$\sin^{-1}\frac{2x}{1+x^2}$, $\tan^{-1}\sqrt{\frac{1-x}{1+x}}$, $(\sin^{-1}x)^2$	Trig substitution first ($x=\tan\theta,\ \cos 2\theta,\ \sin\theta$)	7.6 C, Misc E
$e^x\times$(something)	Rearrange into $e^x[f+f']$; answer $e^x f$	7.6 D
$e^{ax}\sin bx$, $e^{ax}\cos bx$	By parts twice; I comes back; solve for I	7.6 E

If you see...	Do this	See
7. Definite integrals		
Ordinary integrand with limits	Antiderivative, then $F(b) - F(a)$	7.8 A
Substitution inside a definite integral	Change the limits (no back-substitution)	7.9 A
Limits 0 to a; $x \to a - x$ swaps $\sin \leftrightarrow \cos$	P_4, add the two forms: $2I = \int 1\,dx$	7.10 A
$x \cdot f(\sin x)$ from 0 to π, or $\log(1 + \tan x)$	P_4 removes x / simplifies; solve for I	7.10 B
P_4 gives the negative of the integrand	$I = -I \Rightarrow I = 0$	7.10 C
Limits $-a$ to a	Check odd ($= 0$) / even ($= 2\int_0^a$)	7.10 D
Limits 0 to $2a$	P_6: compare $f(2a - x)$ with $f(x)$	7.10 D
$\lvert f(x)\rvert$ in the integrand	Find where $f(x) = 0$; split the interval	7.10 E
$\log \sin x$ or $\log \cos x$ from 0 to $\frac{\pi}{2}$	Use $-\frac{\pi}{2} \log 2$	7.10 B
$\frac{d}{dx} \int_a^x f(t)\,dt$	Answer is $f(x)$ (first fundamental theorem)	7.9

Chapter 8 Application of Integrals

8A JEE basics (extra)

J1. Area between two curves

If $f(x) \geq g(x)$ on $[a, b]$: Area $= \int_a^b \big[f(x) - g(x)\big]\, dx$ (upper curve minus lower curve). Find the intersection points first; they are the limits.

Example: area between $y = x$ and $y = x^2$. They meet at $x = 0, 1$; on $[0, 1]$, $x \geq x^2$.

Area $= \int_0^1 (x - x^2)\, dx = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$ sq. units.

J2. Horizontal strips: use when the curves are easier as $x = \phi(y)$

Area $= \int_c^d \big[x_{\text{right}} - x_{\text{left}}\big]\, dy$.

Example: area between $y^2 = 4x$ and $x = y$. Meet at $y = 0, 4$; right curve $x = y$, left curve $x = \frac{y^2}{4}$.

Area $= \int_0^4 \left(y - \frac{y^2}{4}\right) dy = 8 - \frac{16}{3} = \frac{8}{3}$ sq. units.

J3. Standard areas worth remembering (saves time in MCQs)

Circle $x^2 + y^2 = r^2$	πr^2
Ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$	πab
Parabola $y^2 = 4ax$ and its latus rectum $x = a$	$\frac{8a^2}{3}$
Parabolas $y^2 = 4ax$ and $x^2 = 4ay$	$\frac{16a^2}{3}$
$y = x^2$ and $y = x$	$\frac{1}{6}$
$\lvert x\rvert + \lvert y\rvert = a$ (a square)	$2a^2$

J4. Use symmetry and odd/even functions carefully

For an *area*, every part counts as positive. An odd function has $\int_{-a}^{a} f = 0$, but its area is $2\int_0^a |f|$.

Example: area enclosed by $|x| + |y| = 1$. In the first quadrant it is the triangle under $y = 1 - x$: $\int_0^1 (1 - x)\, dx = \frac{1}{2}$. By symmetry, total area $= 4 \times \frac{1}{2} = 2$ sq. units.

8B Exam tips

Which method? (check in this order)

1. **Sketch the region** and shade it. Mark the curves, lines and axes.
2. **Find the key points**: where the curve meets the axes and the given lines.
3. **Choose strips**: vertical strips ($\int y\,dx$) if the curve is $y = f(x)$; horizontal strips ($\int x\,dy$) if it is $x = \phi(y)$ or bounded by the y-axis.
4. **Use symmetry** (ellipse, circle: $\times 4$; parabola about an axis: $\times 2$).
5. **Split** wherever the curve crosses the axis, and take the absolute value of each part.
6. Integrate, add the parts, and write the answer in **sq. units**.

Common mistakes that lose marks

- No sketch: the figure itself carries marks.
- Reporting a negative area: the part below the x-axis must be taken with modulus.
- Integrating straight across a sign change, e.g. $\int_0^{2\pi} \sin x\,dx = 0$, but the area is 4.
- Forgetting the symmetry factor ($\times 2$ or $\times 4$).
- Mixing limits: x-limits with dy, or y-limits with dx.
- For the ellipse, forgetting the factor $\frac{b}{a}$ in $y = \frac{b}{a}\sqrt{a^2 - x^2}$.
- Leaving out "sq. units".

Smart exam habits and question types

- Draw a neat figure first, label the intersection points, and write "Required area = ...".
- Check the answer against a rough estimate from the figure (e.g. a quadrant of radius 2 is about 3.14).
- **1-mark / MCQ:** area of a quadrant, of a parabola with an axis, by symmetry.
- **5-mark:** area of an ellipse or circle (or part of one), area under $|x + a|$, area of curves crossing the axis.

8C Exercise-wise solution patterns

Exercise 8.1 (4 questions, 3 patterns)

Pattern A: Full ellipse or circle $\Rightarrow$ one quadrant $\times 4$

Spot: closed symmetric curve. **Method:** solve for y, integrate over the first quadrant, multiply by 4.

Solved (Q1): $\frac{x^2}{16} + \frac{y^2}{9} = 1 \Rightarrow y = \frac{3}{4}\sqrt{16 - x^2}$.

$$\text{Area} = 4\int_0^4 \frac{3}{4}\sqrt{16 - x^2}\,dx = 3\left[\frac{x}{2}\sqrt{16 - x^2} + 8\sin^{-1}\frac{x}{4}\right]_0^4 = 3 \cdot 8 \cdot \frac{\pi}{2} = 12\pi \text{ sq. units.}$$

Pattern B: Part of a circle between given lines

Solved (Q3): first quadrant of $x^2 + y^2 = 4$ between $x = 0$ and $x = 2$:

$$\int_0^2 \sqrt{4 - x^2}\,dx = \left[\frac{x}{2}\sqrt{4 - x^2} + 2\sin^{-1}\frac{x}{2}\right]_0^2 = 2 \cdot \frac{\pi}{2} = \pi \quad \textbf{(A)}.$$

Pattern C: Bounded by the y-axis $\Rightarrow$ horizontal strips $\int x\,dy$

Solved (Q4): $y^2 = 4x$, y-axis and $y = 3$. Write $x = \frac{y^2}{4}$: $\int_0^3 \frac{y^2}{4}\,dy = \frac{27}{12} = \frac{9}{4}$ **(B)**.

Q2 is Pattern A (answer 6π).

Miscellaneous Exercise (5 questions, 3 patterns)

Pattern A: Curve above the axis between two ordinates ⇒ direct $\int_a^b y\,dx$

Solved (Q1(ii)): $\displaystyle\int_1^5 x^4\,dx = \left[\frac{x^5}{5}\right]_1^5 = \frac{3125-1}{5} = \frac{3124}{5}$ sq. units.

Pattern B: Curve crosses the x-axis ⇒ split at the crossing, take $|\ |$ of each part

Solved (Q4): $y = x^3$ from $x = -2$ to 1 (negative on $[-2, 0]$):
$\left|\int_{-2}^0 x^3\,dx\right| + \int_0^1 x^3\,dx = |-4| + \frac{1}{4} = \frac{17}{4}$ **(D)**.

Pattern C: Modulus functions ⇒ sketch, remove $|\ |$ piece by piece

Solved (Q5): $y = x|x|$ is x^2 for $x \geq 0$ and $-x^2$ for $x < 0$ (odd function).
The integral from -1 to 1 is 0, but the *area* is $2\int_0^1 x^2\,dx = \frac{2}{3}$ **(C)**.

Q1(i) is Pattern A; Q2 ($|x+3|$, split at -3, area 9) is Pattern C; Q3 ($\sin x$ on $[0, 2\pi]$, area 4) is Pattern B.

8D Pattern spotter

If you see...	Do this	See
Area under a single curve		
Ellipse or circle, full area	One quadrant $\times 4$; use $\int \sqrt{a^2 - x^2}$	8.1 A
Part of a circle between lines	Direct $\int \sqrt{r^2 - x^2}\,dx$ with those limits	8.1 B
Region bounded by the y-axis and $y = c$, $y = d$	Horizontal strips: $\int_c^d x\,dy$	8.1 C
Curve above the x-axis, $x = a$ to $x = b$	$\int_a^b y\,dx$	Misc A
Curve crosses the x-axis inside the interval	Split at the root; add absolute values	Misc B
$\lvert x + a\rvert$, $x\lvert x\rvert$	Sketch; split where the inside is zero	Misc C
Odd function over $[-a, a]$	Integral is 0, but area $= 2\int_0^a \lvert f\rvert$	Misc C
JEE extras		
Two curves	Intersection points, then $\int(\text{upper} - \text{lower})\,dx$	J1
Curves easier as $x = \phi(y)$	$\int(x_{\text{right}} - x_{\text{left}})\,dy$	J2
$\lvert x\rvert + \lvert y\rvert = a$ or other symmetric shapes	One quadrant $\times 4$	J4

Chapter 9 Differential Equations

9A JEE basics (extra)

J1. $\dfrac{dy}{dx} = f(ax+by+c) \Rightarrow$ **put** $v = ax+by+c$

Example: $\dfrac{dy}{dx} = (x+y)^2$. Put $v = x+y$: $\dfrac{dv}{dx} = 1+v^2 \Rightarrow \displaystyle\int \frac{dv}{1+v^2} = \int dx$

$\Rightarrow \tan^{-1}(x+y) = x + C$.

J2. $\dfrac{dy}{dx} = \dfrac{a_1x+b_1y+c_1}{a_2x+b_2y+c_2} \Rightarrow$ **shift the origin, then homogeneous**

Solve $a_1x+b_1y+c_1=0$, $a_2x+b_2y+c_2=0$ to get (h,k); put $x=X+h$, $y=Y+k$. (If $\frac{a_1}{a_2}=\frac{b_1}{b_2}$, use J1 instead.)

Example: $\dfrac{dy}{dx} = \dfrac{x-y+1}{x+y-3}$. The lines meet at $(1,2)$; with $x=X+1$, $y=Y+2$: $\dfrac{dY}{dX} = \dfrac{X-Y}{X+Y}$.

$Y=vX$: $X\dfrac{dv}{dX} = \dfrac{1-2v-v^2}{1+v} \Rightarrow -\frac{1}{2}\log|1-2v-v^2| = \log|X| + C \Rightarrow X^2 - 2XY - Y^2 = C$.

Answer: $(x-1)^2 - 2(x-1)(y-2) - (y-2)^2 = C$.

J3. Bernoulli: $\dfrac{dy}{dx} + Py = Qy^n \Rightarrow$ **divide by** y^n**, put** $z = y^{1-n}$ **(becomes linear)**

Example: $\dfrac{dy}{dx} + \dfrac{y}{x} = y^2$. Divide by y^2 and put $z = y^{-1}$, $\frac{dz}{dx} = -y^{-2}\frac{dy}{dx}$: $\dfrac{dz}{dx} - \dfrac{z}{x} = -1$.

IF $= \dfrac{1}{x}$: $\dfrac{z}{x} = \displaystyle\int -\frac{1}{x}\,dx = -\log|x| + C \Rightarrow \frac{1}{xy} = C - \log|x|$.

J4. Recognise exact differentials

$$x\,dy + y\,dx = d(xy) \qquad \frac{x\,dy - y\,dx}{x^2} = d\left(\frac{y}{x}\right)$$

$$\frac{y\,dx - x\,dy}{y^2} = d\left(\frac{x}{y}\right) \qquad \frac{x\,dy - y\,dx}{x^2+y^2} = d\left(\tan^{-1}\frac{y}{x}\right)$$

$$\frac{x\,dy + y\,dx}{xy} = d(\log xy) \qquad \frac{x\,dx + y\,dy}{x^2+y^2} = d\left(\tfrac{1}{2}\log(x^2+y^2)\right)$$

Example: $x\,dy - y\,dx = (x^2+y^2)\,dx \Rightarrow \dfrac{x\,dy - y\,dx}{x^2+y^2} = dx \Rightarrow \tan^{-1}\dfrac{y}{x} = x + C$.

J5. Forming a differential equation: n **arbitrary constants** $\Rightarrow$ **differentiate** n **times and eliminate**

Example: $y = Ae^{2x} + Be^{-2x}$. Then $y'' = 4Ae^{2x} + 4Be^{-2x} = 4y$, so the equation is $y'' - 4y = 0$ (order 2, as there are 2 constants).

9B Exam tips

Which method? (check in this order)

1. **Can the variables be separated?** $\frac{dy}{dx} = g(x)h(y)$ $\rightarrow$ variable separable.
2. **Is every term the same degree?** $(F(\lambda x, \lambda y) = F(x, y))$ $\rightarrow$ homogeneous: put $y = vx$ (or $x = vy$ if given as $\frac{dx}{dy}$).
3. **Are y and $\frac{dy}{dx}$ both to the power 1, with no $y\frac{dy}{dx}$?** $\rightarrow$ linear in y: make the coefficient of $\frac{dy}{dx}$ equal to 1, then IF $= e^{\int P\,dx}$.
4. **Not linear in y but linear in x?** Rewrite as $\frac{dx}{dy} + P_1 x = Q_1$; IF $= e^{\int P_1\,dy}$.
5. **None of these?** Flip to $\frac{dx}{dy}$, or substitute ($v = x + y$, $x - y = t$), or look for an exact differential.

Order = highest derivative. **Degree** = power of that derivative, defined only when the equation is a polynomial in the derivatives.

Common mistakes that lose marks

- Giving a degree when a derivative sits inside sin, cos, $e^{(\ )}$ or log; the degree is then *not defined*. ($\sin y$ is fine: y is not a derivative.)
- Adding C late. Write $+C$ at the moment you integrate.
- Finding P before dividing to make the coefficient of $\frac{dy}{dx}$ equal to 1.
- Simplifying the IF wrongly: $e^{-\log x} = \frac{1}{x}$, $e^{2\log\sec x} = \sec^2 x$.
- In homogeneous equations, forgetting $\frac{dy}{dx} = v + x\frac{dv}{dx}$.
- Not substituting the given condition to get the particular solution.
- Number of arbitrary constants: general solution = order; particular solution = 0.

Smart exam habits and question types

- State the type first ("This is linear, of the form $\frac{dy}{dx} + Py = Q$"), then write P, Q and IF on separate lines; each earns step marks.
- Check a particular solution by substituting the given point.
- **1-mark / MCQ:** order and degree, number of constants, IF, the substitution for homogeneous equations.
- **3-mark:** variable separable, homogeneous. **5-mark:** linear with a condition, homogeneous particular solution, word problems (growth, slope of a curve).

9C Exercise-wise solution patterns

Exercise 9.1 (12 questions, 2 patterns)

Pattern A: Polynomial in the derivatives ⇒ order and degree both defined

Spot: every derivative appears only with whole-number powers. Functions of y alone (like $\sin y$, y^5) are allowed.

Solved (Q6): $(y''')^2 + (y'')^3 + (y')^4 + y^5 = 0$. Highest derivative y''', raised to power 2: **order 3, degree 2**.

Pattern B: A derivative inside sin, cos, e, log ⇒ degree not defined

Solved (Q1): $\dfrac{d^4y}{dx^4} + \sin(y''') = 0$. Highest derivative is the 4th, so **order 4**; $\sin(y''')$ is not a polynomial, so **degree not defined**.

Pattern A: Q2, 3, 5, 6, 7, 8, 9, 10 (trap: $\sin y$ is allowed), 12. Pattern B: Q1, 4, 11.

Exercise 9.2 (12 questions, 3 patterns)

Pattern A: Verify an explicit solution ⇒ differentiate, substitute, show LHS = RHS

Solved (Q4): $y = \sqrt{1+x^2}$. Then $y' = \dfrac{x}{\sqrt{1+x^2}} = \dfrac{x\sqrt{1+x^2}}{1+x^2} = \dfrac{xy}{1+x^2}$, which is the given equation.

Pattern B: Verify an implicit solution ⇒ differentiate implicitly, solve for y'

Solved (Q9): $x + y = \tan^{-1} y$. Differentiate: $1 + y' = \dfrac{y'}{1+y^2} \Rightarrow y'\left(\dfrac{1}{1+y^2} - 1\right) = 1 \Rightarrow y' = -\dfrac{1+y^2}{y^2}$.

So $y^2y' + y^2 + 1 = 0$. ✓

Pattern C: Number of arbitrary constants

Solved (Q11, Q12): general solution of a 4th-order equation has 4 constants **(D)**; a particular solution has 0 **(D)**.

Pattern A: Q1–6, 10. Pattern B: Q7, 8, 9. Pattern C: Q11, 12.

Exercise 9.3 (23 questions, 4 patterns)

Pattern A: $\dfrac{dy}{dx} = f(x)$ only ⇒ integrate directly (simplify first)

Solved (Q1): $\dfrac{dy}{dx} = \dfrac{1-\cos x}{1+\cos x} = \tan^2\dfrac{x}{2} = \sec^2\dfrac{x}{2} - 1 \Rightarrow y = 2\tan\dfrac{x}{2} - x + C$.

Pattern B: Separate $g(x)$ and $h(y)$, integrate both sides

Solved (Q4): $\sec^2 x \tan y\, dx + \sec^2 y \tan x\, dy = 0 \Rightarrow \dfrac{\sec^2 x}{\tan x}dx = -\dfrac{\sec^2 y}{\tan y}dy$.

$\log|\tan x| = -\log|\tan y| + \log C \Rightarrow \tan x \tan y = C$.

Pattern C: Particular solution $\Rightarrow$ solve, then use the condition to find C

Solved (Q14): $\dfrac{dy}{dx} = y\tan x \Rightarrow \dfrac{dy}{y} = \tan x\,dx \Rightarrow \log|y| = \log|\sec x| + C_1 \Rightarrow y = C\sec x.$
$y = 1$ at $x = 0$ gives $C = 1$, so $y = \sec x$.

Pattern D: Word problems $\Rightarrow$ write the DE from the statement, then separate

Spot: "slope of the tangent", "rate proportional to", "increases continuously".

Solved (Q22): $\dfrac{dN}{dt} = kN \Rightarrow N = N_0e^{kt}$, $N_0 = 10^5$. After 2 hours $N = 1.1\times10^5$: $e^{2k} = 1.1$.
For $N = 2\times10^5$: $e^{kt} = 2 \Rightarrow t = \dfrac{\log 2}{k} = \dfrac{2\log 2}{\log 1.1}$ hours.

Pattern A: Q1, 2, 3, 9, 13. Pattern B: Q4–8, 10, 23. Pattern C: Q11–16. Pattern D: Q17–22.

Exercise 9.4 (17 questions, 3 patterns)

Pattern A: Homogeneous in $\frac{dy}{dx}$ form $\Rightarrow$ put $y = vx$, $\frac{dy}{dx} = v + x\frac{dv}{dx}$

Solved (Q2): $y' = \dfrac{x+y}{x} = 1 + v$. So $v + x\dfrac{dv}{dx} = 1 + v \Rightarrow dv = \dfrac{dx}{x} \Rightarrow v = \log|x| + C \Rightarrow y = x\log|x| + Cx$.

Pattern B: Terms in $\frac{x}{y}$ $\Rightarrow$ write $\frac{dx}{dy}$, put $x = vy$

Solved (Q10): $\left(1 + e^{x/y}\right)dx + e^{x/y}\left(1 - \frac{x}{y}\right)dy = 0$. With $x = vy$, $\frac{dx}{dy} = v + y\frac{dv}{dy}$:
$y\dfrac{dv}{dy} = -\dfrac{v + e^v}{1 + e^v} \Rightarrow \displaystyle\int \frac{1+e^v}{v+e^v}dv = -\int \frac{dy}{y} \Rightarrow \log|v + e^v| = -\log|y| + \log C.$
$\Rightarrow y\left(\frac{x}{y} + e^{x/y}\right) = C$, i.e. $x + ye^{x/y} = C$.

Pattern C: Homogeneous with a condition

Solved (Q12): $x^2dy + (xy + y^2)dx = 0 \Rightarrow \dfrac{dy}{dx} = -(v + v^2)$. Then $x\dfrac{dv}{dx} = -(2v + v^2)$:
$\displaystyle\int \frac{dv}{v(v+2)} = -\int \frac{dx}{x} \Rightarrow \frac{1}{2}\log\left|\frac{v}{v+2}\right| = -\log|x| + C \Rightarrow \frac{y}{y+2x} = \frac{k}{x^2}.$
$y(1) = 1 \Rightarrow k = \frac{1}{3}$: $y + 2x = 3x^2y$.

Pattern A: Q1–9, 17. Pattern B: Q10, 16. Pattern C: Q11–15.

Exercise 9.5 (19 questions, 4 patterns)

Pattern A: Linear in y: $\frac{dy}{dx} + Py = Q$ $\Rightarrow$ $y\cdot\mathbf{IF} = \int Q\cdot\mathbf{IF}\,dx$

Solved (Q6): $x\dfrac{dy}{dx} + 2y = x^2\log x \Rightarrow \dfrac{dy}{dx} + \dfrac{2}{x}y = x\log x$. IF $= e^{2\log x} = x^2$.
$\displaystyle yx^2 = \int x^3\log x\,dx = \frac{x^4}{4}\log x - \frac{x^4}{16} + C \Rightarrow y = \frac{x^2}{16}(4\log x - 1) + \frac{C}{x^2}.$

Pattern B: Linear in x: $\frac{dx}{dy}+P_1x=Q_1 \Rightarrow$ IF $=e^{\int P_1 dy}$

Spot: y appears squared or inside a function, but x is only to power 1.

Solved (Q10): $(x+y)\dfrac{dy}{dx}=1 \Rightarrow \dfrac{dx}{dy}-x=y$. IF $=e^{-y}$.

$xe^{-y}=\displaystyle\int ye^{-y}dy=-ye^{-y}-e^{-y}+C \Rightarrow x=-y-1+Ce^{y}$.

Pattern C: Linear with a condition

Solved (Q13): $\dfrac{dy}{dx}+2y\tan x=\sin x$. IF $=e^{2\log\sec x}=\sec^2 x$.

$y\sec^2 x=\displaystyle\int \sec x\tan x\,dx=\sec x+C \Rightarrow y=\cos x+C\cos^2 x$. $y\left(\frac{\pi}{3}\right)=0 \Rightarrow C=-2$: $y=\cos x-2\cos^2 x$.

Pattern D: Curve from a slope condition

Solved (Q16): slope $=x+y$, through the origin: $\dfrac{dy}{dx}-y=x$, IF $=e^{-x}$.

$ye^{-x}=\displaystyle\int xe^{-x}dx=-xe^{-x}-e^{-x}+C \Rightarrow y=Ce^{x}-x-1$; $(0,0)$ gives $C=1$: $y=e^{x}-x-1$.

Pattern A: Q1–9, 18 (IF $=\frac{1}{x}$). Pattern B: Q10, 11, 12, 19 (IF $=\frac{1}{\sqrt{1-y^2}}$). Pattern C: Q13–15. Pattern D: Q16, 17.

Miscellaneous Exercise (15 questions, 6 patterns)

Pattern A: Order and degree

Solved (Q1): (i) order 2, degree 1; (ii) order 1, degree 3; (iii) $\sin\left(\frac{d^3y}{dx^3}\right)$ present: order 4, degree not defined.

Pattern B: Verify or prove a given solution

Solved (Q2(ii)): $y=e^x(a\cos x+b\sin x)$. Then $y'=y+e^x(-a\sin x+b\cos x)$, so $e^x(-a\sin x+b\cos x)=y'-y$.
$y''=y'+(y'-y)+e^x(-a\cos x-b\sin x)=2y'-y-y=2y'-2y \Rightarrow y''-2y'+2y=0$. ✓

Pattern C: Variable separable (with a condition)

Solved (Q7): $(1+e^{2x})dy+(1+y^2)e^x dx=0 \Rightarrow \dfrac{dy}{1+y^2}=-\dfrac{e^x\,dx}{1+e^{2x}} \Rightarrow \tan^{-1}y=-\tan^{-1}e^x+C$.
$y(0)=1$: $\frac{\pi}{4}=-\frac{\pi}{4}+C \Rightarrow C=\frac{\pi}{2}$: $\tan^{-1}y+\tan^{-1}e^x=\frac{\pi}{2}$.

Pattern D: $\frac{x}{y}$ terms $\Rightarrow$ put $x=vy$

Solved (Q8): $ye^{x/y}dx=(xe^{x/y}+y^2)dy \Rightarrow \dfrac{dx}{dy}=\dfrac{x}{y}+ye^{-x/y}$. With $x=vy$: $y\dfrac{dv}{dy}=ye^{-v}$.
$\Rightarrow e^v dv=dy \Rightarrow e^{x/y}=y+C$.

Pattern E: Linear equations

Solved (Q11): $\dfrac{dy}{dx} + y\cot x = 4x\,\mathrm{cosec}\,x$. IF $= \sin x$: $y\sin x = \int 4x\,dx = 2x^2 + C$.
$y\left(\frac{\pi}{2}\right) = 0 \Rightarrow C = -\frac{\pi^2}{2}$: $y\sin x = 2x^2 - \frac{\pi^2}{2}$.

Pattern F: Substitution suggested by the form

Solved (Q9): $(x-y)(dx+dy) = dx-dy$. Put $t = x-y$ (so $dt = dx-dy$) and note $dx+dy = d(x+y)$:
$t\,d(x+y) = dt \Rightarrow d(x+y) = \dfrac{dt}{t} \Rightarrow x+y = \log|x-y|+C$. $y(0) = -1 \Rightarrow C = -1$: $\log|x-y| = x+y+1$.

Pattern A: Q1. Pattern B: Q2, 3, 5. Pattern C: Q4, 6, 7, 12, 13. Pattern D: Q8. Pattern E: Q10, 11, 14, 15. Pattern F: Q9.

9D Pattern spotter

If you see...	Do this	See
Order and degree		
Equation is a polynomial in the derivatives	Order = highest derivative, degree = its power	9.1 A
Derivative inside sin, cos, e, log	Degree not defined	9.1 B
"How many arbitrary constants?"	General: = order; particular: 0	9.2 C
Verifying solutions		
$y = f(x)$ given	Differentiate, substitute, LHS = RHS	9.2 A
Relation in x and y given	Differentiate implicitly, solve for y'	9.2 B
Constants a, b in the solution	Express them using y, y', then eliminate	Misc B
First-order methods		
$\frac{dy}{dx} = f(x)$ only	Integrate directly (simplify first)	9.3 A
$\frac{dy}{dx} = g(x)h(y)$	Separate, integrate both sides	9.3 B
All terms of the same degree in x, y	$y = vx$	9.4 A
Terms in $\frac{x}{y}$, or $\frac{dx}{dy}$ given	$x = vy$	9.4 B
y and y' to power 1	Linear: IF $= e^{\int P\,dx}$	9.5 A
x to power 1, y messy	Linear in x: IF $= e^{\int P_1\,dy}$	9.5 B
Repeated expression like $x-y$, $x+y$	Substitute it	Misc F
Conditions and applications		
"$y = \ldots$ when $x = \ldots$"	Find C after solving	9.3 C
"Slope of tangent at (x, y) is..."	Write $\frac{dy}{dx} = \ldots$, solve, use the point	9.5 D

If you see…	Do this	See
Growth / interest "continuously"	$\frac{dN}{dt} = kN \Rightarrow N = N_0 e^{kt}$	9.3 D
JEE extras		
$\frac{dy}{dx} = f(ax + by + c)$	$v = ax + by + c$	J1
Ratio of two linear expressions with constants	Shift origin, then homogeneous	J2
$\frac{dy}{dx} + Py = Qy^n$	Bernoulli: $z = y^{1-n}$	J3
$x\,dy \pm y\,dx$ groups	Exact differentials	J4

Chapter 10 Vector Algebra

10A JEE basics (extra)

J1. Scalar triple product $[\vec{a}\ \vec{b}\ \vec{c}] = \vec{a}\cdot(\vec{b}\times\vec{c})$

$$[\vec{a}\ \vec{b}\ \vec{c}] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = \text{volume of the parallelepiped.}\quad \text{Coplanar} \iff [\vec{a}\ \vec{b}\ \vec{c}] = 0.$$

Example: $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b} = 2\hat{i} + 3\hat{j} + 4\hat{k}$, $\vec{c} = 3\hat{i} + 4\hat{j} + 5\hat{k}$:
$1(15-16) - 2(10-12) + 3(8-9) = -1 + 4 - 3 = 0$, so they are coplanar.

J2. Vector triple product

$\vec{a}\times(\vec{b}\times\vec{c}) = (\vec{a}\cdot\vec{c})\,\vec{b} - (\vec{a}\cdot\vec{b})\,\vec{c}$ ("middle one first").

Example: $\vec{a} = \hat{i} + \hat{j}$, $\vec{b} = \hat{j}$, $\vec{c} = \hat{k}$: $(\vec{a}\cdot\vec{c})\vec{b} - (\vec{a}\cdot\vec{b})\vec{c} = 0\cdot\hat{j} - 1\cdot\hat{k} = -\hat{k}$.
Check: $\vec{b}\times\vec{c} = \hat{i}$, and $(\hat{i}+\hat{j})\times\hat{i} = -\hat{k}$. ✓

J3. Useful results

- Lagrange: $|\vec{a}\times\vec{b}|^2 + (\vec{a}\cdot\vec{b})^2 = |\vec{a}|^2|\vec{b}|^2$.
- $|\vec{a}\pm\vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 \pm 2\,\vec{a}\cdot\vec{b}$.
- Vector component of $\vec{a}$ along $\vec{b}$: $\left(\dfrac{\vec{a}\cdot\vec{b}}{|\vec{b}|^2}\right)\vec{b}$. Angle-bisector direction: $\hat{a}+\hat{b}$.
- Area of a parallelogram from its diagonals: $\frac{1}{2}|\vec{d_1}\times\vec{d_2}|$.
- Four points are coplanar $\iff [\overrightarrow{AB}\ \overrightarrow{AC}\ \overrightarrow{AD}] = 0$.

Example: diagonals $\vec{d_1} = 3\hat{i} + \hat{j} - 2\hat{k}$, $\vec{d_2} = \hat{i} - 3\hat{j} + 4\hat{k}$: $\vec{d_1}\times\vec{d_2} = -2\hat{i} - 14\hat{j} - 10\hat{k}$, magnitude $\sqrt{300} = 10\sqrt{3}$.
Area $= \frac{1}{2}\cdot 10\sqrt{3} = 5\sqrt{3}$ sq. units.

10B Exam tips

Which tool? (match the question)

Magnitude / unit vector	$\lvert\vec{a}\rvert = \sqrt{a_1^2 + a_2^2 + a_3^2}$, $\hat{a} = \vec{a}/\lvert\vec{a}\rvert$; vector of magnitude m: $m\hat{a}$
Angle between vectors	$\cos\theta = \dfrac{\vec{a}\cdot\vec{b}}{\lvert\vec{a}\rvert\lvert\vec{b}\rvert}$
Perpendicular	$\vec{a}\cdot\vec{b} = 0$
Parallel / collinear	$\vec{a} = \lambda\vec{b}$ (components proportional), or $\vec{a}\times\vec{b} = \vec{0}$
Projection of $\vec{a}$ on $\vec{b}$	$\dfrac{\vec{a}\cdot\vec{b}}{\lvert\vec{b}\rvert}$
Vector perpendicular to both	$\vec{a}\times\vec{b}$ (unit: $\pm\dfrac{\vec{a}\times\vec{b}}{\lvert\vec{a}\times\vec{b}\rvert}$)
Area of triangle / parallelogram	$\frac{1}{2}\lvert\overrightarrow{AB}\times\overrightarrow{AC}\rvert$ / $\lvert\vec{a}\times\vec{b}\rvert$
Point dividing PQ in $m:n$	Internal $\dfrac{m\vec{q}+n\vec{p}}{m+n}$, external $\dfrac{m\vec{q}-n\vec{p}}{m-n}$

Common mistakes that lose marks

- $\overrightarrow{AB} = \vec{b} - \vec{a}$ (final minus initial), not $\vec{a} - \vec{b}$.
- Projection: divide by $\lvert\vec{b}\rvert$ (the vector you project *on*), not $\lvert\vec{a}\rvert$.
- Cross product: the $\hat{j}$ term has a **minus** sign; and $\vec{b}\times\vec{a} = -\vec{a}\times\vec{b}$.
- Triangle area needs the $\frac{1}{2}$; a unit vector perpendicular to both has **two** answers ($\pm$).
- Direction cosines (divide by the magnitude) are not the same as direction ratios.
- "Collinear" allows opposite directions; collinear vectors need not be equal.

Smart exam habits and question types

- Convert everything to $\hat{i},\hat{j},\hat{k}$ components first. Show the 3×3 determinant for every cross product.
- Check perpendicular answers by computing the dot product.
- **1-mark / MCQ:** unit vector, direction cosines, projection, angle conditions, $\hat{i}\cdot(\hat{j}\times\hat{k})$-type values.
- **2–3 marks:** angle between vectors, λ for perpendicularity, area of a triangle, collinearity, section formula.

10C Exercise-wise solution patterns

Exercise 10.1 (5 questions, 3 patterns)

Pattern A: Represent a displacement graphically

Solved (Q1): draw the North direction, measure 30° towards East, and draw an arrow 4 units long (scale: 1 unit = 10 km). Label the arrow with its magnitude and direction.

Pattern B: Scalar or vector? ⇒ does it need a direction?

Solved (Q2): 10 kg scalar; 2 m north-west **vector**; 40° scalar; 40 watt scalar; 10^{-19} coulomb scalar; 20 m/s^2 **vector** (acceleration).

Pattern C: Types of vectors (coinitial, equal, collinear)

Solved (Q5): (i) $\vec{a}$ and $-\vec{a}$ are collinear: **True**. (ii) Collinear vectors always have equal magnitude: **False**.
(iii) Same magnitude ⇒ collinear: **False**. (iv) Collinear with the same magnitude ⇒ equal: **False** (they may point opposite ways).

Pattern A: Q1. Pattern B: Q2, 3. Pattern C: Q4, 5.

Exercise 10.2 (19 questions, 5 patterns)

Pattern A: Magnitude, unit vector, vector of a given magnitude

Solved (Q10): $5\hat{i}-\hat{j}+2\hat{k}$ has magnitude $\sqrt{30}$. A vector of magnitude 8 along it: $\frac{8}{\sqrt{30}}(5\hat{i}-\hat{j}+2\hat{k}) = \frac{40}{\sqrt{30}}\hat{i}-\frac{8}{\sqrt{30}}\hat{j}+\frac{16}{\sqrt{30}}\hat{k}$.

Pattern B: Components, equality and sums

Solved (Q5): $\overrightarrow{PQ} = (-5-2)\hat{i}+(7-1)\hat{j} = -7\hat{i}+6\hat{j}$. Scalar components $-7,\ 6$; vector components $-7\hat{i},\ 6\hat{j}$.

Pattern C: Direction cosines and collinearity

Solved (Q13): $\overrightarrow{AB} = -2\hat{i}-4\hat{j}+4\hat{k},\ |\overrightarrow{AB}| = 6$. Direction cosines: $-\frac{1}{3},\ -\frac{2}{3},\ \frac{2}{3}$.

Pattern D: Section formula and mid-point

Solved (Q15): $\vec{p} = \hat{i}+2\hat{j}-\hat{k}$, $\vec{q} = -\hat{i}+\hat{j}+\hat{k}$, ratio 2 : 1.
Internal: $\frac{2\vec{q}+\vec{p}}{3} = -\frac{1}{3}\hat{i}+\frac{4}{3}\hat{j}+\frac{1}{3}\hat{k}$. External: $\frac{2\vec{q}-\vec{p}}{2-1} = -3\hat{i}+3\hat{k}$.

Pattern E: Triangle law / right-angled triangle

Solved (Q17): $\overrightarrow{AB} = -\hat{i}+3\hat{j}+5\hat{k}$ ($|\ |^2 = 35$), $\overrightarrow{BC} = -\hat{i}-2\hat{j}-6\hat{k}$ (41), $\overrightarrow{CA} = 2\hat{i}-\hat{j}+\hat{k}$ (6). $35+6 = 41$, so the triangle is right-angled (at A).

Pattern A: Q1, 2, 3, 7, 8, 9, 10. Pattern B: Q4, 5, 6. Pattern C: Q11, 12, 13, 14, 19. Pattern D: Q15, 16. Pattern E: Q17, 18.

Exercise 10.3 (18 questions, 4 patterns)

Pattern A: Angle and projection

Solved (Q2): $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$, $\vec{b} = 3\hat{i} - 2\hat{j} + \hat{k}$: $\vec{a} \cdot \vec{b} = 3 + 4 + 3 = 10$, $|\vec{a}| = |\vec{b}| = \sqrt{14}$. $\cos\theta = \frac{10}{14} = \frac{5}{7} \Rightarrow \theta = \cos^{-1}\frac{5}{7}$.

Pattern B: Expand dot products like algebra (use $\vec{a} \cdot \vec{a} = |\vec{a}|^2$)

Solved (Q13): $\vec{a} + \vec{b} + \vec{c} = \vec{0}$. Dot with itself: $|\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 0$. $3 + 2S = 0 \Rightarrow S = -\frac{3}{2}$.

Pattern C: Perpendicular $\Rightarrow$ dot product $= 0$

Solved (Q10): $\vec{a} + \lambda\vec{b} = (2 - \lambda)\hat{i} + (2 + 2\lambda)\hat{j} + (3 + \lambda)\hat{k}$; dot with $\vec{c} = 3\hat{i} + \hat{j}$: $3(2 - \lambda) + (2 + 2\lambda) = 0 \Rightarrow 8 - \lambda = 0 \Rightarrow \lambda = 8$.

Pattern D: Collinear points / right triangle using vectors

Solved (Q16): $\overrightarrow{AB} = \hat{i} + 4\hat{j} - 4\hat{k}$ and $\overrightarrow{BC} = \hat{i} + 4\hat{j} - 4\hat{k}$. They are equal and share point B, so A, B, C are collinear.

Pattern A: Q1, 2, 3, 4, 15. Pattern B: Q6, 7, 8, 9, 13, 18. Pattern C: Q5, 10, 11, 12, 14. Pattern D: Q16, 17.

Exercise 10.4 (12 questions, 4 patterns)

Pattern A: Cross product and unit vector perpendicular to two vectors

Solved (Q2): $\vec{a} + \vec{b} = 4\hat{i} + 4\hat{j}$, $\vec{a} - \vec{b} = 2\hat{i} + 4\hat{k}$.

$$(\vec{a} + \vec{b}) \times (\vec{a} - \vec{b}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & 4 & 0 \\ 2 & 0 & 4 \end{vmatrix} = 16\hat{i} - 16\hat{j} - 8\hat{k},$$ magnitude 24. Unit vector: $\pm\frac{1}{3}(2\hat{i} - 2\hat{j} - \hat{k})$.

Pattern B: Area of a triangle / parallelogram

Solved (Q9): $\overrightarrow{AB} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\overrightarrow{AC} = 4\hat{j} + 3\hat{k}$. $\overrightarrow{AB} \times \overrightarrow{AC} = -6\hat{i} - 3\hat{j} + 4\hat{k}$, magnitude $\sqrt{61}$. Area $= \frac{\sqrt{61}}{2}$ sq. units.

Pattern C: Cross product $= \vec{0} \iff$ parallel (components proportional)

Solved (Q5): $(2\hat{i} + 6\hat{j} + 27\hat{k}) \times (\hat{i} + \lambda\hat{j} + \mu\hat{k}) = \vec{0} \Rightarrow \frac{2}{1} = \frac{6}{\lambda} = \frac{27}{\mu} \Rightarrow \lambda = 3$, $\mu = \frac{27}{2}$.

Pattern D: Angle conditions and properties

Solved (Q11): $|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}| \sin\theta = 3 \cdot \frac{\sqrt{2}}{3} \sin\theta = 1 \Rightarrow \sin\theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = \frac{\pi}{4}$ **(B)**.

Pattern A: Q1, 2. Pattern B: Q9, 10, 12. Pattern C: Q5, 6, 8. Pattern D: Q3 ($\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$), 4, 7, 11.

Miscellaneous Exercise (19 questions, 6 patterns)

Pattern A: Unit vectors and vectors of a given magnitude

Solved (Q6): $\vec{a}+\vec{b}=3\hat{i}+\hat{j}$, magnitude $\sqrt{10}$. Magnitude 5 along it: $\dfrac{5}{\sqrt{10}}(3\hat{i}+\hat{j})=\dfrac{3\sqrt{10}}{2}\hat{i}+\dfrac{\sqrt{10}}{2}\hat{j}$.

Pattern B: Displacement word problems

Solved (Q3): 4 km west: $-4\hat{i}$. 3 km at 30° east of north: $3\sin 30^\circ\,\hat{i}+3\cos 30^\circ\,\hat{j}=\frac{3}{2}\hat{i}+\frac{3\sqrt{3}}{2}\hat{j}$. Total displacement: $-\frac{5}{2}\hat{i}+\frac{3\sqrt{3}}{2}\hat{j}$.

Pattern C: Collinearity and section formula

Solved (Q8): $\overrightarrow{AB}=4\hat{i}+2\hat{j}+6\hat{k}$, $\overrightarrow{BC}=6\hat{i}+3\hat{j}+9\hat{k}=\frac{3}{2}\overrightarrow{AB}$, so the points are collinear and B divides AC in the ratio $2:3$.

Pattern D: Vector perpendicular to two vectors with an extra condition

Solved (Q12): $\vec{d}=\lambda(\vec{a}\times\vec{b})=\lambda(32\hat{i}-\hat{j}-14\hat{k})$. $\vec{c}\cdot\vec{d}=\lambda(64+1-56)=9\lambda=15\Rightarrow\lambda=\frac{5}{3}$. $\vec{d}=\frac{1}{3}(160\hat{i}-5\hat{j}-70\hat{k})$.

Pattern E: Dot-product conditions and proofs

Solved (Q13): sum $=(2+\lambda)\hat{i}+6\hat{j}-2\hat{k}$. Dot of $\hat{i}+\hat{j}+\hat{k}$ with its unit vector $=1$: $\lambda+6=\sqrt{(2+\lambda)^2+40}\Rightarrow\lambda^2+12\lambda+36=\lambda^2+4\lambda+44\Rightarrow\lambda=1$.

Pattern F: Conceptual MCQs

Solved (Q17): $|\hat{a}+\hat{b}|^2=1+1+2\cos\theta=1\Rightarrow\cos\theta=-\frac{1}{2}\Rightarrow\theta=\frac{2\pi}{3}$ **(D)**.

Pattern A: Q1, 5, 6, 7. Pattern B: Q2, 3. Pattern C: Q8, 9. Pattern D: Q10, 12. Pattern E: Q4, 11, 13, 14, 15. Pattern F: Q16–19.

10D Pattern spotter

If you see...	Do this	See
Basics		
Scalar or vector?	Does it need a direction?	10.1 B
Collinear / equal / coinitial	Same line / same magnitude and direction / same start point	10.1 C
"Unit vector", "vector of magnitude m"	$\hat{a} = \vec{a}/\lvert\vec{a}\rvert$; then $m\hat{a}$	10.2 A
Vector from P to Q	$\overrightarrow{PQ} = \vec{q} - \vec{p}$; scalar components are its coefficients	10.2 B
Direction cosines	Divide each component by the magnitude	10.2 C
Point dividing in $m : n$, mid-point	Section formula (internal / external)	10.2 D
Right-angled triangle from points	Find the three sides; check Pythagoras (or a dot $= 0$)	10.2 E
Dot product		
Angle between vectors	$\cos\theta = \frac{\vec{a}\cdot\vec{b}}{\lvert\vec{a}\rvert\lvert\vec{b}\rvert}$	10.3 A
Projection of $\vec{a}$ on $\vec{b}$	$\frac{\vec{a}\cdot\vec{b}}{\lvert\vec{b}\rvert}$	10.3 A
$(\vec{a}+\vec{b})\cdot(\vec{a}-\vec{b})$, $\lvert\vec{a}+\vec{b}+\vec{c}\rvert$	Expand like algebra; $\vec{a}\cdot\vec{a} = \lvert\vec{a}\rvert^2$	10.3 B
"Perpendicular", find λ	Dot product $= 0$	10.3 C
Three points collinear	$\overrightarrow{AB} = k\,\overrightarrow{BC}$	10.3 D
Cross product		
Vector / unit vector perpendicular to both	$\vec{a}\times\vec{b}$, then $\pm$ unit	10.4 A
Area of triangle or parallelogram	$\frac{1}{2}\lvert\overrightarrow{AB}\times\overrightarrow{AC}\rvert$ or $\lvert\vec{a}\times\vec{b}\rvert$	10.4 B
Cross product $= \vec{0}$	Components proportional	10.4 C
Angles with the axes	$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$	10.4 D
Perpendicular to two vectors with an extra condition	$\vec{d} = \lambda(\vec{a}\times\vec{b})$, use the condition for λ	Misc D
JEE extras		
Volume, coplanar vectors	Scalar triple product (determinant)	J1
$\vec{a}\times(\vec{b}\times\vec{c})$	$(\vec{a}\cdot\vec{c})\vec{b} - (\vec{a}\cdot\vec{b})\vec{c}$	J2
Area from diagonals	$\frac{1}{2}\lvert\vec{d_1}\times\vec{d_2}\rvert$	J3

Chapter 11 Three Dimensional Geometry

11A JEE basics (extra)

J1. The plane (not in the current NCERT book, but needed for JEE)

Through (x_1, y_1, z_1) with normal (a, b, c)	$a(x-x_1)+b(y-y_1)+c(z-z_1)=0$
Vector (normal) form	$\vec{r}\cdot\vec{n}=d$
Intercept form	$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$
Through 3 points	$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ x_2-x_1 & y_2-y_1 & z_2-z_1 \\ x_3-x_1 & y_3-y_1 & z_3-z_1 \end{vmatrix}=0$

Example: plane through $(1,1,1)$ with normal $\hat{i}+2\hat{j}+3\hat{k}$: $(x-1)+2(y-1)+3(z-1)=0 \Rightarrow x+2y+3z=6$.

J2. Distances and angles with planes

Point to plane: $\dfrac{|ax_1+by_1+cz_1-d|}{\sqrt{a^2+b^2+c^2}}$. Parallel planes: $\dfrac{|d_1-d_2|}{\sqrt{a^2+b^2+c^2}}$.

Between planes: $\cos\theta=\dfrac{|\vec{n}_1\cdot\vec{n}_2|}{|\vec{n}_1||\vec{n}_2|}$. Line and plane: $\sin\phi=\dfrac{|\vec{b}\cdot\vec{n}|}{|\vec{b}||\vec{n}|}$.

Example: distance of $(2,1,-1)$ from $x-2y+4z=9$: $\dfrac{|2-2-4-9|}{\sqrt{1+4+16}}=\dfrac{13}{\sqrt{21}}$.

J3. Foot of perpendicular and image of a point in a line

Write a general point of the line using λ; make (that point $-$ given point) perpendicular to the line's direction; solve for λ. Image $=2\times$ foot $-$ point.

Example: from $A(1,6,3)$ to $\frac{x}{1}=\frac{y-1}{2}=\frac{z-2}{3}$. General point $P(\lambda,\,2\lambda+1,\,3\lambda+2)$.
$(\lambda-1)\cdot1+(2\lambda-5)\cdot2+(3\lambda-1)\cdot3=14\lambda-14=0\Rightarrow\lambda=1$: foot $(1,3,5)$, image $(1,0,7)$, distance $\sqrt{13}$.

J4. Do two lines intersect? (coplanarity test)

Lines $\vec{r}=\vec{a}_1+\lambda\vec{b}_1$ and $\vec{r}=\vec{a}_2+\mu\vec{b}_2$ are coplanar $\iff$ $(\vec{a}_2-\vec{a}_1)\cdot(\vec{b}_1\times\vec{b}_2)=0$. If they are coplanar and not parallel, they intersect (shortest distance 0).

11B Exam tips

Which method? (check in this order)

1. **Put every line in standard form** $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$ (coefficients of x, y, z must be $+1$). Read off the point and the direction ratios.
2. **Angles given?** Direction cosines are the cosines of the angles. **DRs given?** Divide by $\sqrt{a^2+b^2+c^2}$.
3. **Perpendicular** $\Rightarrow a_1a_2 + b_1b_2 + c_1c_2 = 0$. **Parallel** $\Rightarrow$ DRs proportional.
4. **Angle between lines:** $\cos\theta = \frac{|a_1a_2+b_1b_2+c_1c_2|}{\sqrt{\cdot}\sqrt{\cdot}}$.
5. **Shortest distance:** skew lines $d = \frac{|(\vec{a}_2-\vec{a}_1)\cdot(\vec{b}_1\times\vec{b}_2)|}{|\vec{b}_1\times\vec{b}_2|}$; parallel lines $d = \frac{|\vec{b}\times(\vec{a}_2-\vec{a}_1)|}{|\vec{b}|}$.
6. **Line perpendicular to two lines:** direction $= \vec{b}_1 \times \vec{b}_2$.

Common mistakes that lose marks

- Reading DRs from a non-standard form: $\frac{1-x}{3}$ means DR -3; $\frac{7y-14}{2p} = \frac{y-2}{2p/7}$.
- Mixing up direction cosines ($l^2+m^2+n^2=1$) and direction ratios (any multiple).
- Using the skew-lines formula for parallel lines: $\vec{b}_1 \times \vec{b}_2 = \vec{0}$ there.
- Vector form: $\vec{r} =$ (point) $+\lambda$(direction), not the other way round.
- Forgetting the modulus: the angle between lines is acute, and the distance is positive.

Smart exam habits and question types

- Write $\vec{a}_1$, $\vec{b}_1$, $\vec{a}_2$, $\vec{b}_2$ clearly before any formula; show the determinant for $\vec{b}_1 \times \vec{b}_2$.
- **1-mark / MCQ:** direction cosines, condition for perpendicular lines, vector $\leftrightarrow$ cartesian form.
- **3–5 marks:** shortest distance between two lines, angle between lines, finding k or p for perpendicular lines, line perpendicular to two given lines.

11C Exercise-wise solution patterns

Exercise 11.1 (5 questions, 3 patterns)

Pattern A: Direction cosines from angles

Solved (Q1): angles $90°, 135°, 45°$: $l = \cos 90° = 0$, $m = \cos 135° = -\frac{1}{\sqrt{2}}$, $n = \cos 45° = \frac{1}{\sqrt{2}}$.

Pattern B: Direction cosines from direction ratios or two points

Solved (Q3): DRs $-18, 12, -4$; $\sqrt{324+144+16} = 22$. DCs: $-\frac{9}{11}$, $\frac{6}{11}$, $-\frac{2}{11}$.

Pattern C: Collinear points $\Rightarrow$ DRs of AB and BC proportional

Solved (Q4): $A(2,3,4)$, $B(-1,-2,1)$, $C(5,8,7)$. DRs of AB: $-3,-5,-3$; of BC: $6,10,6 = -2(-3,-5,-3)$.
Proportional, with B common: collinear.

Pattern A: Q1, 2 ($l = m = n = \pm\frac{1}{\sqrt{3}}$). Pattern B: Q3, 5. Pattern C: Q4.

Exercise 11.2 (15 questions, 4 patterns)

Pattern A: Perpendicular / parallel lines via direction ratios

Solved (Q10): write in standard form: $\frac{x-1}{-3}=\frac{y-2}{2p/7}=\frac{z-3}{2}$ and $\frac{x-1}{-3p/7}=\frac{y-5}{1}=\frac{z-6}{-5}$.
Perpendicular: $(-3)\left(-\frac{3p}{7}\right)+\frac{2p}{7}\cdot 1+2(-5)=0\Rightarrow\frac{11p}{7}=10\Rightarrow p=\frac{70}{11}$.

Pattern B: Equation of a line (vector ↔ cartesian)

Solved (Q6): through $(-2,4,-5)$, parallel to the line with DRs $3,5,6$: $\dfrac{x+2}{3}=\dfrac{y-4}{5}=\dfrac{z+5}{6}$.
Vector form: $\vec{r}=(-2\hat{i}+4\hat{j}-5\hat{k})+\lambda(3\hat{i}+5\hat{j}+6\hat{k})$.

Pattern C: Angle between two lines

Solved (Q9(ii)): DRs $(2,2,1)$ and $(4,1,8)$: $\cos\theta=\dfrac{8+2+8}{3\cdot 9}=\dfrac{2}{3}\Rightarrow\theta=\cos^{-1}\dfrac{2}{3}$.

Pattern D: Shortest distance between skew lines

Solved (Q12): $\vec{a}_1=\hat{i}+2\hat{j}+\hat{k}$, $\vec{b}_1=\hat{i}-\hat{j}+\hat{k}$; $\vec{a}_2=2\hat{i}-\hat{j}-\hat{k}$, $\vec{b}_2=2\hat{i}+\hat{j}+2\hat{k}$.
$\vec{b}_1\times\vec{b}_2=-3\hat{i}+3\hat{k}$ ($|\ |=3\sqrt{2}$); $\vec{a}_2-\vec{a}_1=\hat{i}-3\hat{j}-2\hat{k}$; dot $=-3-6=-9$.
$d=\dfrac{9}{3\sqrt{2}}=\dfrac{3\sqrt{2}}{2}$.

Pattern A: Q1, 2, 3, 10, 11. Pattern B: Q4–7. Pattern C: Q8, 9. Pattern D: Q12–15 (Q15: first rewrite each line as point + parameter × direction).

Miscellaneous Exercise (5 questions, 3 patterns)

Pattern A: Angle / perpendicular condition with letters

Solved (Q3): DRs $(-3,2k,2)$ and $(3k,1,-5)$: $-9k+2k-10=0\Rightarrow k=-\frac{10}{7}$.

Pattern B: Line with a special direction (axis, or perpendicular to two lines)

Solved (Q5): direction $=\vec{b}_1\times\vec{b}_2=(3\hat{i}-16\hat{j}+7\hat{k})\times(3\hat{i}+8\hat{j}-5\hat{k})=24\hat{i}+36\hat{j}+72\hat{k}=12(2\hat{i}+3\hat{j}+6\hat{k})$.
$\vec{r}=(\hat{i}+2\hat{j}-4\hat{k})+\lambda(2\hat{i}+3\hat{j}+6\hat{k})$.

Pattern C: Shortest distance (check parallel or skew first)

Solved (Q4): $\vec{b}_1=\hat{i}-2\hat{j}+2\hat{k}$, $\vec{b}_2=3\hat{i}-2\hat{j}-2\hat{k}$ (not parallel, so skew).
$\vec{b}_1\times\vec{b}_2=8\hat{i}+8\hat{j}+4\hat{k}$ ($|\ |=12$); $\vec{a}_2-\vec{a}_1=-10\hat{i}-2\hat{j}-3\hat{k}$; $d=\dfrac{|-80-16-12|}{12}=9$.

Pattern A: Q1 (dot product $=0$, so 90°), 3. Pattern B: Q2 ($\frac{x}{1}=\frac{y}{0}=\frac{z}{0}$), 5. Pattern C: Q4.

11D Pattern spotter

If you see...	Do this	See
Direction cosines and ratios		
Angles with the axes given	DCs = cosines of the angles	11.1 A
Equal angles with the axes	$l = m = n = \pm\frac{1}{\sqrt{3}}$	11.1 A
DRs or two points given	Divide by the length	11.1 B
Three points collinear?	DRs of AB, BC proportional	11.1 C
Lines		
Line in an odd form $(\frac{1-x}{3}, \frac{7y-14}{2p})$	Rewrite with x, y, z coefficients +1	11.2 A
"Perpendicular", find p or k	$a_1a_2 + b_1b_2 + c_1c_2 = 0$	11.2 A
Through a point, parallel to a vector/line	$\vec{r} = \vec{a} + \lambda\vec{b}$ / cartesian form	11.2 B
Angle between lines	$\cos\theta = \frac{\lvert\vec{b}_1\cdot\vec{b}_2\rvert}{\lvert\vec{b}_1\rvert\lvert\vec{b}_2\rvert}$	11.2 C
Line perpendicular to two lines	Direction $\vec{b}_1 \times \vec{b}_2$	Misc B
Shortest distance		
Two skew lines	$\frac{\lvert(\vec{a}_2-\vec{a}_1)\cdot(\vec{b}_1\times\vec{b}_2)\rvert}{\lvert\vec{b}_1\times\vec{b}_2\rvert}$	11.2 D
Parallel lines	$\frac{\lvert\vec{b}\times(\vec{a}_2-\vec{a}_1)\rvert}{\lvert\vec{b}\rvert}$	11B
Lines given with t, s mixed in	Rewrite as point + parameter × direction	11.2 D
JEE extras		
Plane through a point with a normal	$a(x-x_1)+b(y-y_1)+c(z-z_1)=0$	J1
Distance from a point to a plane	$\frac{\lvert ax_1+by_1+cz_1-d\rvert}{\sqrt{a^2+b^2+c^2}}$	J2
Foot of perpendicular / image in a line	General point with λ, dot with direction $= 0$	J3
Do two lines intersect?	$(\vec{a}_2 - \vec{a}_1)\cdot(\vec{b}_1 \times \vec{b}_2) = 0$	J4

Chapter 12 Linear Programming

12A JEE basics (extra)

J1. Formulating a word problem

Define the variables, write the objective, turn each resource limit into a constraint, and add $x, y \geq 0$.

Example: toy A needs 2 h machine time and 1 h polishing; toy B needs 1 h and 1 h. Available: 10 h machine, 6 h polishing. Profit: Rs 5 (A), Rs 3 (B).
Maximise $Z = 5x + 3y$ subject to $2x + y \leq 10$, $x + y \leq 6$, $x, y \geq 0$.
Corners: $(0,0) \to 0$, $(5,0) \to 25$, $(4,2) \to 26$, $(0,6) \to 18$. Maximum Rs 26 at $(4,2)$.

J2. Iso-profit (iso-cost) line method

Draw $Z = k$ for some k and slide it parallel to itself. For a maximum, the last corner the line touches is optimal (for a minimum, the first). If the line ends up lying along an edge, every point of that edge is optimal.

J3. Quick facts

- A **bounded** feasible region always has both a maximum and a minimum, at corner points.
- An **unbounded** region may have no maximum (or no minimum); always do the half-plane check.
- If the constraints contradict each other, there is **no feasible region** and so no optimal value.
- The feasible region is always convex.

12B Exam tips

Corner point method (follow every time)

1. Draw each constraint line using its intercepts. Shade the correct side by testing $(0,0)$.
2. Find the feasible region (common to all, with $x, y \geq 0$). Note whether it is bounded.
3. List every corner point; solve pairs of line equations for the intersection corners.
4. Make a table of Z at each corner.
5. **Bounded:** largest value is the max, smallest is the min.
6. **Unbounded:** if m is the smallest value, check whether $Z < m$ has any point in common with the region. If not, m is the minimum. (For a maximum M, check $Z > M$.)

Common mistakes that lose marks

- Shading the wrong side of a line ($\geq$ vs $\leq$).
- Forgetting $x \geq 0$, $y \geq 0$, or missing a corner point.
- Arithmetic slips when solving two lines for a corner. Substitute back to check.
- Declaring a max/min on an unbounded region without the half-plane check.
- With two corners giving the same optimum, not writing "at every point on the segment joining them".

Smart exam habits and question types

- Draw the graph to scale with labelled lines, shade the feasible region, and mark all corners.
- Always present the corner-point table; it carries most of the marks.
- **1-mark / MCQ:** corner points of a given region, value of Z at a corner, bounded or unbounded.
- **5-mark:** solve a graphical LPP (often a case-based word problem).

12C Exercise-wise solution patterns

Exercise 12.1 (10 questions, 5 patterns)

Pattern A: Bounded region ⇒ corner point table

Solved (Q3): Maximise $Z = 5x + 3y$; $3x + 5y \le 15$, $5x + 2y \le 10$, $x, y \ge 0$.
Intersection: from $5x + 2y = 10$, $y = \frac{10-5x}{2}$; then $3x + \frac{5(10-5x)}{2} = 15 \Rightarrow x = \frac{20}{19}$, $y = \frac{45}{19}$.

Corner	$(0,0)$	$(2,0)$	$(0,3)$	$\left(\frac{20}{19}, \frac{45}{19}\right)$
Z	0	10	9	$\frac{235}{19}$

Maximum $Z = \frac{235}{19}$ at $\left(\frac{20}{19}, \frac{45}{19}\right)$.

Pattern B: Unbounded region, minimise ⇒ corner table plus half-plane check

Solved (Q4): Minimise $Z = 3x + 5y$; $x + 3y \ge 3$, $x + y \ge 2$, $x, y \ge 0$.
Corners: $(3,0) \to 9$, $\left(\frac{3}{2}, \frac{1}{2}\right) \to 7$, $(0,2) \to 10$. Smallest is 7.
$3x + 5y < 7$ has no point in common with the region, so the minimum is 7 at $\left(\frac{3}{2}, \frac{1}{2}\right)$.

Pattern C: Two corners give the same optimum ⇒ the whole edge is optimal

Solved (Q6): Minimise $Z = x + 2y$; $2x + y \ge 3$, $x + 2y \ge 6$, $x, y \ge 0$.
Corners: $(6,0) \to 6$ and $(0,3) \to 6$. $x + 2y < 6$ has no common point with the region.
Minimum 6 at every point of the segment joining $(6,0)$ and $(0,3)$, i.e. at more than two points.

Pattern D: Unbounded region, maximise ⇒ there may be no maximum

Solved (Q9): Maximise $Z = -x + 2y$; $x \ge 3$, $x + y \ge 5$, $x + 2y \ge 6$, $y \ge 0$.
Corners: $(6,0) \to -6$, $(4,1) \to -2$, $(3,2) \to 1$. But $-x + 2y > 1$ has points in the region (e.g. $(3,3)$ gives 3).
So Z has **no maximum**.

Pattern E: No feasible region

Solved (Q10): $x - y \le -1$ means $y \ge x + 1$; $-x + y \le 0$ means $y \le x$. Both cannot hold, so there is no feasible region and no maximum.

Pattern A: Q1, 2, 3, 5, 7 (max on an edge), 8. Pattern B: Q4. Pattern C: Q6, and the edge answers in Q7, 8. Pattern D: Q9. Pattern E: Q10.

12D Pattern spotter

If you see...	Do this	See
Setting up		
Word problem	Variables, objective Z, one constraint per resource, $x, y \geq 0$	J1
Shading a constraint	Test $(0, 0)$ in the inequality	12B
Corner from two lines	Solve the two equations together	12.1 A
Reading the answer		
Region closed on all sides	Corner table; largest/smallest Z	12.1 A
Region open, asked to minimise	Corner table, then check $Z < m$	12.1 B
Two corners give the same optimum	Every point on the segment joining them	12.1 C
Region open, asked to maximise	Check $Z > M$; often no maximum	12.1 D
Constraints contradict each other	No feasible region, no optimum	12.1 E
Quick visual method	Slide the iso-profit line $Z = k$	J2

Chapter 13 Probability

13A JEE basics (extra)

J1. Binomial distribution (repeated independent trials)

$P(X=r)=\binom{n}{r}p^r q^{n-r}$, $q=1-p$. Mean $=np$, variance $=npq$.

Example: 4 fair coins; $P(\text{at least 3 heads})=\binom{4}{3}\frac{1}{16}+\binom{4}{4}\frac{1}{16}=\frac{5}{16}$.

J2. Mean and variance of a random variable

$E(X)=\sum x_i p_i$, $\quad \text{Var}(X)=E(X^2)-[E(X)]^2$.

Example: $X=$ number of heads in 2 tosses: values $0,1,2$ with probabilities $\frac{1}{4},\frac{1}{2},\frac{1}{4}$.
$E(X)=1$, $E(X^2)=\frac{1}{2}+1=\frac{3}{2}$, $\text{Var}(X)=\frac{3}{2}-1=\frac{1}{2}$.

J3. "Who wins first?" (alternate turns)

If A starts and each turn succeeds with probability p ($q=1-p$): $P(A \text{ wins})=p+q^2p+q^4p+\cdots=\dfrac{p}{1-q^2}$.

Example: A and B throw a die alternately; first to get a 6 wins: $P(A)=\dfrac{1/6}{1-25/36}=\dfrac{6}{11}$.

J4. "At least one" with several independent events

$P(\text{at least one})=1-P(A')P(B')P(C')\cdots$

Example: independent probabilities $\frac{1}{2},\frac{1}{3},\frac{1}{4}$: $1-\frac{1}{2}\cdot\frac{2}{3}\cdot\frac{3}{4}=\frac{3}{4}$.

13B Exam tips

Which method? (read the question's key words)

1. **"given that", "if it is known that"** → conditional: $P(E|F)=\frac{P(E\cap F)}{P(F)}$, or count in the reduced sample space F.
2. **"one after another", "without replacement"** → multiplication rule: $P(A)P(B|A)P(C|AB)$.
3. **"independent"** → $P(A\cap B)=P(A)P(B)$; test it by comparing both sides.
4. **"at least one"** → $1-P(\text{none})$.
5. **A result is observed; find the chance it came from a particular source** → Bayes' theorem.
6. **Overall probability of a result from several sources** → total probability $\sum P(E_i)P(A|E_i)$.

Common mistakes that lose marks

- Confusing $P(A|B)$ with $P(B|A)$: the event after the bar is the one you know happened.
- Treating mutually exclusive as independent. (Mutually exclusive events with non-zero probabilities are never independent.)
- Multiplying probabilities of events that are not independent.
- In Bayes, choosing events $E_1,\ldots,E_n$ that do not cover all cases or that overlap.
- Forgetting that the second draw changes after "without replacement" or "transferred".

Smart exam habits and question types

- Name the events in words first ("E_1: bag I is chosen; A: ball is red").
- For Bayes, draw a small tree or table of $P(E_i)$ and $P(A|E_i)$ before substituting.
- **1-mark / MCQ:** $P(A|B)$ from given numbers, independence checks, concept questions like $A \subset B$.
- **3–5 marks:** Bayes' theorem word problems (very common), at-least-one questions, conditional probability with dice and coins.

13C Exercise-wise solution patterns

Exercise 13.1 (17 questions, 4 patterns)

Pattern A: Numbers given $\Rightarrow$ use $P(E|F) = \frac{P(E\cap F)}{P(F)}$ and the addition rule

Solved (Q5): $P(A) = \frac{6}{11}$, $P(B) = \frac{5}{11}$, $P(A \cup B) = \frac{7}{11}$.
$P(A \cap B) = \frac{6}{11} + \frac{5}{11} - \frac{7}{11} = \frac{4}{11}$; $P(A|B) = \frac{4/11}{5/11} = \frac{4}{5}$; $P(B|A) = \frac{4/11}{6/11} = \frac{2}{3}$.

Pattern B: An experiment (coins, dice, family) $\Rightarrow$ count in the reduced sample space

Solved (Q10(a)): the black die shows 5, so only the red die varies (6 outcomes). Sum > 9 needs red $= 5$ or 6.
$P = \frac{2}{6} = \frac{1}{3}$.

Pattern C: Counting from a table of data

Solved (Q13): multiple-choice questions $= 500 + 400 = 900$, of which 500 are easy. $P(\text{easy} \mid \text{MCQ}) = \frac{500}{900} = \frac{5}{9}$.

Pattern D: Concept MCQs

Solved (Q17): $P(A|B) = P(B|A) \Rightarrow \frac{P(A\cap B)}{P(B)} = \frac{P(A\cap B)}{P(A)} \Rightarrow P(A) = P(B)$ **(D)**.

Pattern A: Q1–5. Pattern B: Q6–12, 14, 15. Pattern C: Q13. Pattern D: Q16 (not defined when $P(B) = 0$), 17.

Exercise 13.2 (18 questions, 4 patterns)

Pattern A: Independence $\Rightarrow P(A \cap B) = P(A)P(B)$ (use it or test it)

Solved (Q7): $P(A) = \frac{1}{2}$, $P(A \cup B) = \frac{3}{5}$, $P(B) = p$.
(i) Mutually exclusive: $\frac{3}{5} = \frac{1}{2} + p \Rightarrow p = \frac{1}{10}$. (ii) Independent: $\frac{3}{5} = \frac{1}{2} + p - \frac{p}{2} \Rightarrow p = \frac{1}{5}$.

Pattern B: Successive draws without replacement $\Rightarrow$ multiplication rule

Solved (Q3): 12 good and 3 bad oranges; approved if all 3 drawn are good:
$\frac{12}{15} \cdot \frac{11}{14} \cdot \frac{10}{13} = \frac{44}{91}$.

Pattern C: "At least one" / "exactly one"

Solved (Q14): $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$, independent.
Solved: $1 - \frac{1}{2} \cdot \frac{2}{3} = \frac{2}{3}$. Exactly one: $\frac{1}{2} \cdot \frac{2}{3} + \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{2}$.

Pattern D: Percentages / set data $\Rightarrow$ addition rule and conditional formula

Solved (Q16): $P(H) = 0.6$, $P(E) = 0.4$, $P(H \cap E) = 0.2$.
(a) Neither: $1 - (0.6 + 0.4 - 0.2) = 0.2$. (b) $P(E|H) = \frac{0.2}{0.6} = \frac{1}{3}$. (c) $P(H|E) = \frac{0.2}{0.4} = \frac{1}{2}$.

Pattern A: Q1, 4, 5, 6, 7, 8, 10, 11, 15, 18. Pattern B: Q2, 3, 13, 17. Pattern C: Q12, 14. Pattern D: Q9, 16.

Exercise 13.3 (14 questions, 3 patterns)

Pattern A: Bayes' theorem $\Rightarrow P(E_i|A) = \dfrac{P(E_i)P(A|E_i)}{\sum_j P(E_j)P(A|E_j)}$

Spot: a result is observed, and you are asked which source/cause it came from.
Solved (Q4): E_1: knows ($\frac{3}{4}$), E_2: guesses ($\frac{1}{4}$); A: answers correctly. $P(A|E_1) = 1$, $P(A|E_2) = \frac{1}{4}$.

$$P(E_1|A) = \frac{\frac{3}{4} \cdot 1}{\frac{3}{4} \cdot 1 + \frac{1}{4} \cdot \frac{1}{4}} = \frac{12/16}{13/16} = \frac{12}{13}.$$

Pattern B: Total probability (no "reverse" question)

Solved (Q1): first ball red ($\frac{1}{2}$): urn becomes 7 red, 5 black; first black ($\frac{1}{2}$): 5 red, 7 black.
$P(\text{second red}) = \frac{1}{2} \cdot \frac{7}{12} + \frac{1}{2} \cdot \frac{5}{12} = \frac{1}{2}$.

Pattern C: Bayes where $P(A|E_i)$ needs its own calculation

Solved (Q10): E_1: die shows 5 or 6 ($\frac{1}{3}$), then 3 tosses: $P(\text{exactly one head}) = \frac{3}{8}$.
E_2: die shows 1–4 ($\frac{2}{3}$), then 1 toss: $P(\text{one head}) = \frac{1}{2}$.

$$P(E_2|A) = \frac{\frac{2}{3} \cdot \frac{1}{2}}{\frac{2}{3} \cdot \frac{1}{2} + \frac{1}{3} \cdot \frac{3}{8}} = \frac{1/3}{11/24} = \frac{8}{11}.$$

Pattern A: Q2–9, 11, 13. Pattern B: Q1. Pattern C: Q10, 12. Q14 is a concept MCQ: $A \subset B \Rightarrow P(A|B) = \frac{P(A)}{P(B)} \geq P(A)$ **(C)**.

Miscellaneous Exercise (13 questions, 5 patterns)

Pattern A: Concept questions on $P(B|A)$

Solved (Q1): (i) $A \subset B \Rightarrow A \cap B = A \Rightarrow P(B|A) = 1$. (ii) $A \cap B = \phi \Rightarrow P(B|A) = 0$.

Pattern B: Reduced sample space / counting

Solved (Q2(i)): at least one male: $\{MM, MF, FM\}$, so $P(\text{both male}) = \frac{1}{3}$.
Solved (Q5): a leap year has 52 weeks + 2 extra days; of the 7 possible pairs, 2 contain a Tuesday: $\frac{2}{7}$.

Pattern C: Bayes' theorem (including transfer problems)

Solved (Q10): transfer red ($\frac{3}{7}$): bag II has 5 red, 5 black; transfer black ($\frac{4}{7}$): 4 red, 6 black.

$$P(\text{black transferred} \mid \text{red drawn}) = \frac{\frac{4}{7} \cdot \frac{4}{10}}{\frac{3}{7} \cdot \frac{5}{10} + \frac{4}{7} \cdot \frac{4}{10}} = \frac{16}{31}.$$

Pattern D: Probabilities given as data ⇒ read the words carefully

Solved (Q9): $P(A \cap B) = 0.15$ and $P(B \text{ alone}) = 0.15$, so $P(B) = 0.30$.
(i) $P(A|B) = \frac{0.15}{0.30} = 0.5$. (ii) $P(A \text{ alone}) = 0.2 - 0.15 = 0.05$.

Pattern E: Repeated independent trials

Solved (Q4): each person is right-handed with $p = 0.9$; at most 6 of 10:
$P = \sum_{r=0}^{6} \binom{10}{r} (0.9)^r (0.1)^{10-r}$.

Pattern A: Q1, 11, 12, 13. Pattern B: Q2, 5, 8 ($ad = 1$, $bc = 0$: $\frac{1}{4} \cdot \frac{3}{4} = \frac{3}{16}$). Pattern C: Q3, 6, 7, 10. Pattern D: Q9. Pattern E: Q4.

13D Pattern spotter

If you see...	Do this	See
Conditional probability		
$P(A)$, $P(B)$, $P(A \cup B)$ given	Find $P(A \cap B)$ by the addition rule, then divide	13.1 A
"Given that" with coins / dice / children	List the reduced sample space and count	13.1 B
Table of counts	Ratio of counts inside the given group	13.1 C
$A \subset B$, $A \cap B = \phi$	$P(B\|A) = 1$ / 0	Misc A
Independence and multiplication		
"Independent"	$P(A \cap B) = P(A)P(B)$	13.2 A
"Mutually exclusive"	$P(A \cap B) = 0$	13.2 A
Draws one after another, no replacement	Multiply, updating the counts	13.2 B
"At least one"	$1 - P(\text{none})$	13.2 C
Percentages of overlapping groups	Addition rule, then conditional formula	13.2 D
Bayes' theorem		
Result seen, find the source	Bayes: $\frac{P(E_i)P(A\|E_i)}{\sum P(E_j)P(A\|E_j)}$	13.3 A
Overall chance of a result	Total probability $\sum P(E_i)P(A\|E_i)$	13.3 B
Each source has its own small experiment	Work out each $P(A\|E_i)$ first	13.3 C
Ball transferred between bags	Sources = colour transferred	Misc C

If you see…	Do this	See
JEE extras		
n repeated trials, r successes	Binomial $\binom{n}{r}p^r q^{n-r}$	J1
Mean / variance of X	$\sum xp$, $E(X^2) - E(X)^2$	J2
Players take turns	$\frac{p}{1-q^2}$ for the first player	J3